Machine Learning Forecasts of US CPI Inflation: Evidence from the 2026 FRED-MD Vintage

Duy Phan

(2802376)

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Thesis commission:

Mingxuan Song (Supervisor)

Dr. Siem Jan Koopman (Co-reader)

Contents

Abstract

This thesis compares classical time series models against Machine Learning (ML) models and identifies key predictors for forecasting US CPI inflation within a rich data environment using the FRED-MD dataset. The empirical analysis leads to the following key findings. First, ML methods, including Random Forest, Ridge regression, and adaptive LASSO, applied to a large updated dataset, achieve higher predictive accuracy than traditional time-series benchmarks across medium- to long-term forecasting horizons, particularly during more volatile inflation periods. Among the competing models, Random Forest demonstrates superior prediction performance and maintains a stable variable importance across horizons. Second, standalone tree-based models perform better than their hybrid specifications, suggesting that feature selection before model estimation does not necessarily enhance model accuracy. Lastly, apart from inflation lags, which are the most influential and reliable predictive factors, other variables relevant to the housing market, the labor market, interest and exchange rates, and money and credit are also among the most important predictors for forecasting US CPI inflation. Altogether, these results contribute to macroeconomic time series forecasting and highlight the benefits of ML methods, together with big data, in achieving higher prediction accuracy.

Keywords: Inflation Forecasting, Machine Learning, Big Data, Random Forest.

1 Introduction

Inflation is a sustained increase in the general price level of goods and services. It is considered one of the most important key economic indicators. The future inflation rate can heavily affect households, economic agents, authorities, and policymakers in making optimal economic decisions and investment strategies over time. According to Zhukova and Flerova (2022) and Rondinelli and Zizza (2020), high inflation rates weaken economic growth, reduce wages and savings, and raise production costs, whereas low inflation rates indicate lower demand in the economy, as indicated by lower spending and consumption and investment project durations. Central banks usually rely on inflation forecasts to implement effective economic policies and to anchor the expected inflation rate, thereby stabilizing the economy. Therefore, producing accurate inflation forecasts plays a crucial role in economic decision-making (Szyszko 2015). Inflation forecasting is a complicated task. This mainly stems from the presence of high correlation, nonlinearity, and high dimensionality among the macroeconomic and financial variables, as well as potential structural breaks in the data, which challenge traditional forecasting techniques and the formulation of a forecasting equation (Aras and Lisboa 2022, Doojav and Narmandakh 2025). However, the combination of recent advancements in ML models and the availability of big datasets appears to be a potential alternative solution to these issues.

There is disagreement among previous inflation forecasting studies about whether multivariate time-series models outperform univariate ones. In particular, Atkeson et al. (2001) evaluate the usefulness of the Phillips curve-based models for inflation forecasting based on real economic activity, but conclude that the search for other augmented Phillips-curve models should be abandoned due to their lower forecast accuracies compared to naive models. The Phillips curve with different specifications yields more reliable and stable inflation forecasts than univariate forecasting models (Stock and Watson 1999). However, their work is limited by relying solely on linear models, which are prone to overfitting, and a single 12-month forecasting horizon was used to evaluate the models. As a result, existing research on inflation forecasting is inadequate for modeling the complex relationship between covariates and inflation, leading to inaccurate predictions that can undermine the effectiveness of the central bank's policy. More importantly, the benefits of ML models combined with large datasets have been scarcely explored in these studies on macroeconomic forecasting.

There has been a rising focus on the recent development of ML models and easy accessibility to large datasets, and economists and researchers have considered them potential alternatives to classical time-series models to enhance inflation forecasting accuracy (Doojav and Narmandakh 2025). Several studies have been conducted to verify this finding. One of them is Nakamura (2005), who claims that the performance of the AR model is normally worse than that of neural networks in forecasting US inflation one to two quarters ahead. According to Garcia et al. (2017), the best short-term inflation forecasts in the case of the Brazilian economy are made by shrinkage models including LASSO and adaptive LASSO combined with numerous regressors. Numerous studies show a preference for ML techniques in data-rich environments. A comprehensive analysis by Medeiros et al. (2021) confirms the superior performance of Random Forest (RF) models in US inflation prediction and supports the idea that incorporating ML methods with large datasets beats the univariate benchmark models. However, the dominance of RF is not universal across countries and the different model sets, according to Naghi et al. (2024), who find that other tree-based models, particularly gradient boosted trees (GBTs), have better forecast performance than RF when forecasting CPI inflation in Canada with an extended set of ML models. Doojav and Narmandakh (2025) argue that ML forecasting methods combined with big data, namely, XGBoost and Random Forest, yield more accurate long-term Mongolian inflation forecasts than AR and FAAR benchmarks, which only excel at short forecasting horizons. Another key result of Huang et al. (2025) shows that the ML shrinkage models Ridge Regression and Elastic Net are superior for inflation forecasting in China compared to conventional time-series benchmarks at medium and long forecast horizons during periods of fluctuating inflation.

A hybrid ML forecasting approach is a combination of two different methodologies, where the first hybrid part uses one model to select variables, and the second part includes a chosen ML model being estimated on this compressed set of features to make an inflation forecast. Its goal is to test whether the model performance is determined mainly by the variable selection or nonlinear patterns in the data. Medeiros et al. (2021) considers some specifications such as adaLASSO-RF and RF-OLS, while Araujo and Gaglianone (2023) extends the set of six hybrid models, namely, AdaLASSO-RF, AdaLASSO-XGBoost, AdaLASSO-OLS, RF-OLS, RF-XGBoost, and RF-AdaLASSO. In terms of identifying the key inflation drivers, the set of Predictors vary among studies. Medeiros et al. (2021) highlights the predictive group importance of exchange and interest rates, labor market, prices, housing, and lagged inflation for US inflation at each horizon during the full out-of-sample period. For predicting China's PPI and CPI inflation, Huang et al. (2025) emphasized that stock, money and credit, price, and AR terms are the four most crucial predictor groups under both full out-of-sample and two subsamples of calm and fluctuating inflation periods. In addition, the set of best predictors of Mongolian inflation varies across forecasting horizons according to Doojav and Narmandakh (2025). Wages and prices contribute the most to short-term forecasts. In contrast, long-term inflation has predictors related to interest and exchange rates, money and credit, expenditure and revenue, and output and income. More importantly, the global oil price is also mentioned as one of the most important predictors in forecasting medium-term Mongolian inflation (Doojav and Narmandakh 2025).

Given the lack of literature regarding evaluating the predictive performance of the GAS model along with the common time-series and ML models on a newly updated 2026 US dataset, the dispute of previous studies about the effectiveness of ML models in producing more accurate inflation forecasts than traditional benchmarks, whether a hybrid ML approach improves the model's accuracy, and whether Random Forest remains the dominant forecasting method in different evaluation samples, horizons, and model sets, this thesis aims to close these gaps. First, it compares and evaluates time-series models, including the Autoregressive (AR) model, Random Walk (RW), AORW, and the Generalized Autoregressive Score model (GAS) against Machine Learning (ML) models such as LASSO, Ridge Regression (RR), adaptive LASSO (adaLASSO), Elastic Net (ElNet), Random Forest (RF), XGBoost, and hybrid ML models (adaLASSO-RF, adaLASSO-XGBoost) in US CPI inflation prediction across different forecast horizons during both the full out-of-sample period and two subsamples of more- and less-volatile inflation periods by using the most recently updated large FRED-MD monthly macroeconomic US dataset. Second, the predictive performance of ML hybrid specifications is examined to assess the effectiveness of feature-selection techniques before model estimation in enhancing the quality of US CPI inflation forecasts. Finally, I also identify the features with the greatest explanatory power for the US inflation forecasting exercise, based on the best-performing ML models.

This study contributes to the inflation-forecasting literature, which can be summarized as follows. Firstly, it provides a comprehensive comparison and assessment of ML and classical benchmark models for US CPI inflation forecasting across different horizons under both the full-sample and the subsample analyses. Secondly, this work can better identify the most informative predictors, potentially serving as key drivers of US inflation dynamics, and determine the contribution of each selected predictor to the prediction problem in terms of variable importance under the three mentioned evaluation samples. The final contribution is to test whether feature selection before model estimation improves the inflation-forecasting performance of ML models. These findings can further enhance inflation forecasting accuracy and, therefore, help central banks achieve optimal monetary policies that stabilize their inflation rates.

However, there are limitations to this study. Firstly, my ML forecasting pipeline may have poor generalization to other countries due to differences in each country's economy, inflation dynamics, and predictor sets, which lead to distinct volatility patterns in inflation rates. Secondly, the findings are based on a fixed set of 12 models rather than an extensive set of various forecasting methods. Finally, adaptive LASSO is the only shrinkage-based model employed to preselect features in the first step of the hybrid ML approach.

The rest of my thesis is constructed as follows. Sections 2 and 3 describe the methodology, which includes an inflation-forecasting equation, traditional econometric methods, and ML models. Section 4 introduces the data and hyperparameter tuning for ML models. Section 5 provides empirical results in comparing and evaluating the forecasts of competing models and variable group importance analysis. Section 6 is the conclusion. The appendix A provides additional stationary transformation tables in the data processing steps.

2 Methodology

Following the methodology of Medeiros et al. (2021), a direct multi-month-ahead inflation forecasting framework is considered to forecast the future US CPI inflation rate π_{t+h} at month $t+h$ using the set of information available at time t, namely $\mathbf{X}_t$. The main reason for this choice is the robustness of the direct forecasting approach when the model is misspecified, which often happens for the alternative iterated or recursive forecasting method (Marcellino et al. 2006, Huang et al. 2025, Doojav and Narmandakh 2025). This can be demonstrated by the following equation

$$\pi_{t+h} = F_h(\mathbf{X}_t) + u_{t+h}, \qquad h = 1, 3, 6, 9, 12 \quad \text{and} \quad t = 1, \ldots, T,$$

where the $(n \times 1)$ predictor vector $\mathbf{X_t} = (x_{1t}, \ldots, x_{nt})'$ contains the US CPI inflation lags and a set of many macroeconomic variables up to time t, which is mapped to the forecasted target variable π_{t+h} by the function $F_h(\mathbf{X}_t)$. 1-, 3-, 6-, 9-, and 12-month-ahead forecasts are the forecasting horizons h, and the error term u_{t+h} has a mean of zero. Each forecasting horizon h yields a distinct mapping function. Moreover, $F_h(\mathbf{X}_t)$ is considered to be in one of the following model specifications: it can be a benchmark time series model, or a ML model, or a hybrid ML model. The estimated form of the direct forecasting equation is as follows

$$\widehat{\pi}_{t+h|t} = \widehat{F}_{h,\,t-R_h+1:t}(\mathbf{X}_t)\,,$$

where the size of the window R_h is set to $R_h = N_0 - h$, computed once for each forecasting horizon by subtracting horizon h from the number of in-sample observations N_0 before the out-of-sample date in January 1990, and this deviates from the lag-based window size of Medeiros et al. (2021). The data from time $t - R_h + 1$ to t is used to estimate the target function $\widehat{F}_{h,\,t-R_h+1:t}$. Forecasting US CPI inflation is constructed using different models $F_h(\mathbf{X}_t)$, including benchmark models (AR, Random Walk, AORW, GAS) and supervised Machine Learning models, including regularized linear regression models (Ridge Regression, LASSO, Elastic Net, adaptive LASSO), tree-based models (Random Forest (RF), and extreme gradient boosting (XGBoost)), and additional hybrid ML models (adaLASSO-RF, adaLASSO-XGBoost) to verify the importance of preselected features in enhancing model forecasting ability. I also standardize the set of variables as predictor inputs for shrinkage-based models before estimation. Tree-based models, by contrast, are applied to unscaled features (Araujo and Gaglianone 2023). In the case of hybrid ML models, scaled predictors are used only in the first step of adaptive LASSO feature preselection, which returns the indices of relevant variables. RF and XGBoost are then fitted on these selected unscaled features. These models are described in the section below.

3 Models

3.1 Benchmark Models

This subsection considers a set of benchmark models for US CPI inflation forecasts. π_t denotes the monthly inflation rate at time t. The time-series benchmarks in this thesis, namely RW, AR, and AORW, are standard choices in inflation-forecasting studies. Huang et al. (2025) employ AR(4) and AORW as benchmark models. Medeiros et al. (2021) use RW, AR, and UCSV models as the three benchmarks for forecasting inflation in the United States. Araujo and Gaglianone (2023) include AORW and RW with six other benchmark models for forecasting Brazilian inflation, while

Doojav and Narmandakh (2025) use AR(p) and FAAR as two benchmarks. The Generalized Autoregressive Score (GAS) model is considered together with the three models above because there is a lack of studies that compare and evaluate this score-driven model alongside ML methods for US CPI inflation in a data-rich setting across the full out-of-sample and subsample periods. Finally, RW is used as a baseline model to compute the metric ratios in Section 5 for model comparison, following Medeiros et al. (2021).

3.1.1 Random Walk Model (RW)

The first benchmark model is the Random Walk model (RW). For the forecast horizons $h = 1, 3, 6, 9, 12$, the inflation forecast of RW for period $t + h$ made at time t is $\widehat{\pi}_{t+h|t} = \pi_t$, which is the last observed value of inflation (Medeiros et al. 2021).

3.1.2 Autoregressive Model (AR(p))

The second benchmark model is the autoregressive model of order p, namely, AR(p). The lag order p is determined by the Bayesian Information Criterion (BIC) for each window (Medeiros et al. 2021), with the defined search space $p \in \{1, 2, 3, 4\}$ (Doojav and Narmandakh 2025). This is different from the case of Huang et al. (2025), who fix the inflation lag length at four. The OLS method is used to estimate the parameters $\beta_{i,h}$. The forecast equation is as follows:

$$\hat{\pi}_{t+h|t} = \hat{\beta}_{0,h} + \hat{\beta}_{1,h}\,\pi_t + \cdots + \hat{\beta}_{p,h}\,\pi_{t-p+1},$$

where $\hat{\pi}_{t+h|t}$ represents the h-month-ahead forecasted inflation given the information available up to date t. The parameters $\hat{\beta}_{i,h}$ for $i = 0, 1, \ldots, p$ are OLS estimates and $\pi_t, \ldots, \pi_{t-p+1}$ are the lagged values of inflation. The assumption of linearity is imposed by this time-series model between the target variable and lagged inflation.

3.1.3 AORW Model

The third benchmark is the AORW forecasting model, which is essentially a variant of the Random Walk forecast (Atkeson et al. 2001). In particular, averaging inflation over the last 12 months is used to obtain the forecast. Therefore, the AORW forecast is defined as follows:

$$\pi^h_{t+h|t} = \pi^{12}_t = \frac{1}{12}\left(\pi_t + \pi_{t-1} + \cdots + \pi_{t-11}\right)$$

This benchmark univariate model is also employed in the studies of forecasting inflation in China (Huang et al. 2025) , and in the case study of Brazil (Araujo and Gaglianone 2023).

3.1.4 Generalized Autoregressive Score Model (GAS(1,1))

The final benchmark is the GAS(1,1) model, an observation-driven time-series model introduced by Creal et al. (2013) and Harvey (2013). The main idea behind this model is to update the parameter over time. This is achieved by the scaled score of the likelihood with a chosen scaling matrix, where the steepest ascent direction for enhancing the model's local fit is determined by the score. Applying this to the US CPI inflation forecasting framework, we formulate the model specification as follows. Let π_t be the variable of interest at time t; the time-dependent parameter here is set as the mean f_t of inflation; the variance is assumed to be constant σ^2, and $\theta = (\omega, A, B, \sigma^2)$

is a vector of static parameters. The reason behind this setup is that this study concentrates on predicting the average inflation level in $\%$ instead of inflation volatility; therefore, the conditional mean f_t is chosen to move over time, the variance is fixed, and the lag orders are fixed at $p = 1$ and $q = 1$ to have a parsimonious model. The distribution of π_t given the available information $\mathcal{F}_{t-1}$ is Gaussian with a conditional mean and constant variance.

$$\pi_t|\mathcal{F}_{t-1} \sim \mathcal{N}(f_t, \sigma^2), \quad \pi_t = \log(P_t) - \log(P_{t-1}), \quad t = 1, \ldots, T$$

The observation density and the corresponding log-likelihood function are therefore given by

$$\pi_t \sim p(\pi_t|f_t, \mathcal{F}_{t-1}, \theta)$$

$$\ln p(\pi_t|f_t, \mathcal{F}_{t-1}, \theta) = \ell_t = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(\pi_t - f_t)^2}{2\sigma^2}$$

The updating equation for the GAS(1,1) model is

$$f_{t+1} = \omega + As_t + Bf_t$$

where ω is a constant, A and B are coefficients. The recursion is initialized at the in-sample mean of each window, $f_1 = \bar{\pi}$. Then the algorithm updates f_t to the next time period $t + 1$ with the scaled score, where the derivative of the log-likelihood with respect to the mean f_t is the score ∇_t.

$$\nabla_t = \frac{\partial \ell_t}{\partial f_t} = \frac{\pi_t - f_t}{\sigma^2}$$

The scaling matrix S_t is set equal to the inverse of the information matrix.

$$S_t = I_{t|t-1}^{-1} = \sigma^2, \quad I_{t|t-1} = \mathbf{E}_{t-1}[\nabla_t^2] = \frac{1}{\sigma^2}$$

The scaled score is the driving mechanism of the GAS(1,1) model; in this case, it is defined as the forecast error.

$$s_t = S_t \cdot \nabla_t = \sigma^2 \cdot \frac{\pi_t - f_t}{\sigma^2} = \pi_t - f_t$$

It is easy to see that the choice of the inverse information matrix of $S_t = \sigma^2$ simplifies the raw score ∇_t to the simple forecast error term $\pi_t - f_t$, avoiding the chance of dividing by a tiny value of variance σ^2, which can potentially lead to numerical instability. The maximum likelihood approach is used to estimate the static parameter vector.

$$\hat{\theta} = (\hat{\omega}, \hat{A}, \hat{B}, \hat{\sigma}^2) = \arg\max_{\theta} \sum_{t=1}^{T} \ell_t$$

Replacing s_t with the obtained forecast error, we have the recursion for the GAS(1,1) model:

$$f_{t+1} = \omega + A(\pi_t - f_t) + Bf_t = \omega + A\pi_t + (B - A)f_t$$

This can be interpreted as an error-correction average inflation model, with a current forecast of average inflation f_t for each month t. This current forecast is then corrected and updated via the forecast error, and this continues until the end of the sample. The update equation of GAS(1,1) model is as follows:

$$f_{t+1} = \omega + As_t + Bf_t, \quad s_t = \pi_t - f_t$$

Since the score $s_t = \pi_t - f_t$ is the forecast error, the update equation only remains valid if the observed inflation π_t at month t is available. However, the last inflation observation is π_{t-1} because the in-sample window stops at month $t-1$, before the test date t. In that case, the future scaled score s_{t+j} is replaced with the conditional expectation

$$E_{t-1}[s_{t+j}] = E_{t-1}[\pi_{t+j} - f_{t+j}] = 0 \quad \text{for all } j \geq 0, \quad \text{where} \quad \pi_{t+j} \mid \mathcal{F}_{t+j-1} \sim \mathcal{N}(f_{t+j}, \sigma^2).$$

Therefore, the score term $A \cdot s_t$ or forecast error disappears for subsequent forecasts. The recursion for the GAS(1,1) model therefore reduces to only $\omega + B \cdot f_t$. The update equation for the forecast at month t given the last observed inflation π_{t-1}:

$$\hat{f}_{t|t-1} = \omega + A(\pi_{t-1} - f_{t-1}) + Bf_{t-1}$$

The scaled score term is removed because no realized inflation rates are available from month t onward for subsequent inflation forecasts.

$$\hat{f}_{t+j|t-1} = \omega + B\, \hat{f}_{t+j-1|t-1}, \quad j = 1, \ldots, h,$$

where the recursion starts from $\hat{f}_{t|t-1}$. Unfolding this recursion yields the final closed-form expression for the h-step-ahead inflation forecasting framework.

$$\hat{f}_{t+h|t-1} = \omega \, \frac{1 - B^h}{1 - B} + B^h \, \hat{f}_{t|t-1} = \hat{\pi}_{t+h|t-1},$$

where $B \neq 1$. Moreover, if $|B| < 1$, then the inflation forecasts of GAS(1,1) converge to unconditional mean $\frac{\omega}{(1-B)}$ as the horizon h increases. Thus, future inflation forecasts of the GAS model are obtained conditional on the inflation observations available through month $t-1$.

3.2 Shrinkage Models

Regularized regression models are introduced to address overfitting due to a high number of variables by including an additional penalty function in linear regression. These methods, therefore, reduce the impact of irrelevant variables by shrinking their coefficients $\beta_{h,i}$ toward zero, and to exactly zero in some cases. In this section, Ridge Regression, LASSO, adaptive LASSO, and Elastic Net models are considered (Medeiros et al. 2021, Huang et al. 2025, Doojav and Narmandakh 2025, Aras and Lisboa 2022). The format of the penalty function varies according to these models, and the minimization problem is as follows

$$\hat{\beta}_h = \arg\min_{\beta_h} \left[\sum_{t=1}^{T-h} (\pi_{t+h} - \beta_h' \mathbf{X}_t)^2 + \sum_{i=1}^{N} p(\beta_{h,i}; \lambda, \omega_i) \right],$$

where λ is a penalty parameter that determines the strength of the penalty function $p(\beta_{h,i}; \lambda, \omega_i)$ and $\omega_i > 0$ is a weight parameter. T and N represent the number of observations and predictors, respectively. The mapping function specification $F_h(\cdot)$ is defined as a linear regression model:

$$F_h(\mathbf{X}_t) = \beta_h' \mathbf{X}_t = \beta_{0,h} + \sum_{i=1}^{N} \beta_{h,i} x_{it}$$

3.2.1 Ridge Regression (RR)

Ridge Regression was introduced by Hoerl and Kennard (1970). The objective of RR is to set regression coefficients close to zero but never exactly to zero for predictors with tiny contributions to the target variable. To do this, a linear regression model is penalized with an L2-norm penalty, indicated by the equation below

$$\sum_{i=1}^{N} p(\beta_{h,i}, \lambda, \omega_i) := \lambda \sum_{i=1}^{N} \beta_{h,i}^2$$

RR has feasible computation (Medeiros et al. 2021) and handles a potentially high correlation between predictors (Aras and Lisboa 2022). However, RR cannot guarantee obtaining a parsimonious model, as all variables are retained within the model. Another interesting point is that RR is a special case of Elastic Net when setting $\alpha = 1$ in the penalty equation (1) (Araujo and Gaglianone 2023).

3.2.2 The Least Absolute Shrinkage and Selection Operator (LASSO)

LASSO model, introduced by Tibshirani (1996), is an alternative to RR. The main idea of LASSO is that coefficients $\beta_{h,i}$ of irrelevant predictors are shrunk exactly to zero by imposing an L1-norm penalty on the regression model.

$$\sum_{i=1}^{N} p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^{N} |\beta_{h,i}|$$

The shrinkage nature of LASSO defines it as a feature selection technique (Aras and Lisboa 2022). LASSO is also a special case of the Elastic Net in the case of $\alpha = 0$ in the ElNet penalty equation (1) (Araujo and Gaglianone 2023).

3.2.3 Adaptive LASSO (adaLASSO)

Adaptive LASSO is an enhanced version of LASSO and was developed by Zou (2006) to attain more consistent variable selection by using an additional weighting parameter ω_i with the same L1 penalty term as LASSO. Therefore, adaLASSO with a two-step estimation procedure can select variables more effectively using the prior information (Araujo and Gaglianone 2023). The penalty function of adaLASSO is demonstrated as follows

$$\sum_{i=1}^{N} p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^{N} \omega_i |\beta_{h,i}|$$

where $\omega_i = |\tilde{\beta}_{h,i}|^{-1}$ shows the weights, and $\tilde{\beta}_{h,i}$ is the respective estimated LASSO coefficient in the first step. Following Medeiros et al. (2021) and Huang et al. (2025), I set the weights $\omega_i = \frac{1}{|\tilde{\beta}_{h,i}| + \frac{1}{\sqrt{T}}}$ as the first-step LASSO sets some coefficients to exactly zero, therefore, adding the term $\frac{1}{\sqrt{T}}$ to the denominator prevents division by zero $\frac{1}{|\tilde{\beta}_{h,i}|}$ and undefined weights for variables.

3.2.4 Elastic Net (ElNet)

Elastic Net model, introduced by Zou and Hastie (2005), is a combined penalized specification of both Ridge and LASSO models for linear regression. It is able to shrink coefficient variables close to zero and set them exactly to zero for unimportant variables at the same time through balancing the weights of L1- and L2-norm penalty terms with the additional parameter $\alpha \in [0, 1]$. The penalty function is given by:

$$\sum_{i=1}^{N} p(\beta_{h,i}; \lambda, \omega_i) := \alpha\lambda \sum_{i=1}^{N} \beta_{h,i}^2 + (1-\alpha)\lambda \sum_{i=1}^{N} |\beta_{h,i}| \tag{1}$$

This integration helps Elastic Net obtain useful properties, namely, shrinkage and sparsity offered by the two penalty terms (Huang et al. 2025, Aras and Lisboa 2022).

3.3 Tree-based ML Models

Nonlinear and complex patterns of the FRED-MD dataset are difficulties that linear models often struggle to capture correctly; therefore, Random Forest and XGBoost are additional tree-based ML methods considered besides the above-mentioned shrinkage models to handle these challenges effectively.

3.3.1 Random Forest Models (RF)

According to Araujo and Gaglianone (2023) and Garcia et al. (2017), a regression tree recursively divides the predictor space $\mathbf{X}$ via a binary partition approach and is a nonparametric method. A single regression tree has the shape of a binary decision tree and usually faces issues such as overfitting and high-variance forecasts (Aras and Lisboa 2022). To address these problems, Random Forest models, as proposed by Breiman (2001), are an ensemble model that includes multiple decision trees. In particular, RF relies on bootstrap aggregating (bagging) and considers a random subset of features at each split for each tree to ensure the trees are as different as possible. Therefore, the random forest model can balance model interpretability with forecasting accuracy (Doojav and Narmandakh 2025). However, running the RF algorithm is computationally expensive and time-consuming (Dewi and Chen 2019). The following equation represents the RF model

$$\pi_{t+h} = \sum_{k=1}^{K} c_k \, I\{x_t \in R_k\} \tag{2}$$

$$I(x_t \in R_k) = \begin{cases} 1, & x_t \in R_k, \\ 0, & \text{otherwise.} \end{cases}$$

where $I(x_t \in R_k)$ is an indicator function, c_k is the constant corresponding to the k^{th} region. The constant c_k is obtained by minimizing the sum of squared errors of the regression equation (2), and this results in $\hat{c}_k = \text{mean}(\pi_{t+h}^k)$ as the sample mean of realized inflation in the region R_k, which is denoted as π_{t+h}^k. K is the number of leaves of a generic tree, while K_b represents the number of terminal nodes of a specific tree b within an ensemble. R_k denotes the partition region corresponding to the k^{th} terminal node of the tree with $k = 1, 2, \ldots, K$. The procedure can be described as follows. First, B bootstrap samples are obtained from the original data, indexed by $b = 1, \ldots, B$. Then each tree is estimated using a random subset of the predictors considered at each node. Finally, the final RF inflation forecast is achieved by averaging the inflation forecasts across B trees (Huang et al. 2025, Medeiros et al. 2021).

$$\hat{\pi}_{t+h} = \frac{1}{B} \sum_{b=1}^{B} \left[\sum_{k=1}^{K_b} \hat{c}_k I\{x_t \in R_k\} \right]$$

3.3.2 Extreme Gradient Boosting (XGBoost)

Apart from the Random Forest model, another tree-based ensemble method is XGBoost, introduced by Chen and Guestrin (2016), which builds on the gradient boosting framework. Its core idea is that each newly added tree is trained sequentially to reduce the errors made by the previous ones. Therefore, the loss function is minimized by this iterative process. The following equations can mathematically represent the XGBoost model:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i),$$

where $\hat{y}_i$ denotes the final forecasted inflation value, K represents the number of trees, and $f_k(x_i)$ stands for the prediction of the k^{th} tree. The objective function of XGBoost is indicated as follows:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

where the first term, the differentiable loss function $l(\hat{y}_i, y_i)$, measures the discrepancy between the true value y_i and the predicted value $\hat{y}_i$, and it determines how well the model fits the data, while the regularization term

$$\Omega(f) = \gamma J + \frac{1}{2}\lambda\|w\|^2$$

is used to govern model complexity and reduce potential model overfitting. These are achieved by the hyperparameters λ to penalize the vector of leaf weights $\|w\|$ and γ to penalize the tree complexity, and J stands for the number of leaves of a tree. XGBoost is considered in this analysis for the following reasons. This tree-based model reduces overfitting by including a penalty term, and training speed is accelerated through parallel processing. However, the performance of XGBoost is determined largely by a set of hyperparameters (Aras and Lisboa 2022).

3.4 Hybrid ML Models

A hybrid inflation-forecasting method is employed to test whether selecting features in advance improves the predictive performance of ML models. Following Medeiros et al. (2021), the procedure is conducted as follows:

1. In each rolling window, the algorithm first estimates the adaptive LASSO model and keeps the predictors with non-zero coefficients. A safety condition is set to retain the top two features with the largest absolute coefficient values when adaLASSO selects fewer than two variables.

2. The tree-based models, including RF and XGBoost, are then estimated and tuned on the set of predictors selected by adaLASSO to make inflation forecasts.

This yields two hybrid ML methods: adaLASSO-RF and adaLASSO-XGBoost. Although more combinations of shrinkage and tree-based models could be considered, these two hybrid models are sufficient to examine whether selecting variables before estimation improves the ML forecast accuracy.

3.5 Statistical Test

3.5.1 Diebold-Mariano (DM) Test

The Diebold-Mariano (DM) test (Diebold and Mariano 2002) is used in this thesis to examine whether the discrepancy in forecasting accuracy between two models is statistically significant. In particular, this test is formulated using the loss differential procedure, which involves choosing an appropriate loss function for forecast errors and checking whether the average of the loss difference is significantly different from zero. In this case, I apply a two-sided DM test to compare the model with the best performance against each benchmark at every forecasting horizon over the full out-of-sample period. This is conducted in the following steps:

1. Define the forecast errors as $\hat{u}_{it} = \hat{\pi}_{it} - \pi_t, \quad i = 1, 2 \quad$ and $t = 1, \ldots, T, \quad$ where $i = 1$ represents the best-performing model (RF) and $i = 2$ each of the benchmark models (RW, AR, GAS, AORW).

2. The loss differential is $d_t = g(u_{1t}) - g(u_{2t})$, where $g(\cdot)$ is the loss function. The squared loss $g(u_{it}) = u_{it}^2$ and the absolute loss $g(u_{it}) = |u_{it}|$ are considered in this case.

3. Under the null hypothesis, inflation forecasts of both models are equally accurate, while the alternative hypothesis means the forecasts of the two models are not equally accurate:

$$H_0 : E(d_t) = 0 \quad \forall t \quad \text{vs} \quad H_1 : E(d_t) \neq 0$$

4. The DM test statistic under H_0 is defined as

$$DM = \frac{\bar{d}}{\hat{\sigma}_{\bar{d}}}$$

where $\bar{d} = \frac{1}{T} \sum_{t=1}^{T} d_t$ is the average loss difference and the standard error $\hat{\sigma}_{\bar{d}} = \sqrt{\widehat{\text{Var}}(\bar{d})}$ is computed based on a HAC estimator that has autocovariances up to lag $h - 1$ (Medeiros et al. 2021). I use significance levels of 1%, 5%, and 10%.

5. Small-sample adjustment of Harvey et al. (1997) is considered in this statistical test due to the finite number of out-of-sample inflation forecasts, and the adjusted test statistic is compared with a Student's t-distribution with $T - 1$ degrees of freedom.

4 Data Description and Hyperparameter Tuning

4.1 Data

Table 1: Descriptive statistics of the US CPI inflation

	Mean	Sd	Max	Min
Low Volatility Period: 1990–1999	0.242	0.163	0.946	0.000
High Volatility Period: 2000–2025	0.211	0.302	1.367	-1.786

The input data I used is extracted from the FRED-MD database, which is a monthly panel of US macroeconomic time series, compiled and updated by McCracken and Ng (2016). Other studies

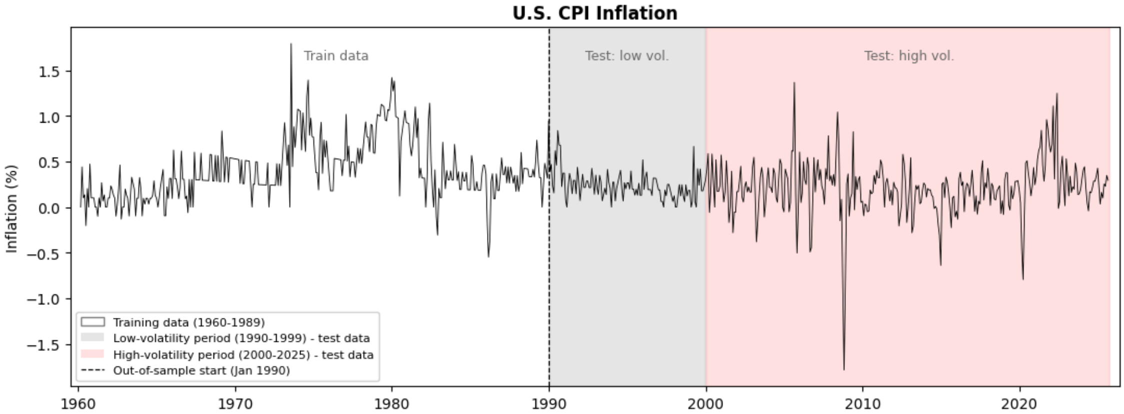


Figure 1: Time Series Plot for Monthly US CPI Inflation over the Full Sample Period

such as Medeiros et al. (2021), Giannone et al. (2021), and Granziera and Sekhposyan (2019) use this dataset for macroeconomic time-series forecasting. The vintage data is the May 2026 version. The data sample is fixed to span from January 1960 to September 2025, with a total of 789 observations. The beginning date of the sample follows Medeiros et al. (2021), and I decided to end the sample in September 2025 to have a complete dataset at the cost of the last seven months of the vintage because CPI and 20 other series have missing observations in October 2025. The 2025 US government shutdown prevented the BLS from collecting survey data to publish an October CPI (U.S. Bureau of Labor Statistics 2026). I then conduct the following data preprocessing steps. Firstly, I retain only 119 series with complete observations over the sample period (Medeiros et al. 2021). Second, I apply the stationary transformation for each time series, as listed in Appendix A. Following the approach of McCracken and Ng (2016), I identify outliers among all predictor series by flagging any transformed observations whose distance from the sample median exceeds ten interquartile ranges. These outliers are set to missing and replaced with the most recent valid non-outlier value in the same series to avoid potential look-ahead bias. Finally, a total panel of 480 predictors is formed by considering four lags of every predictor and four autoregressive terms of the inflation target (Medeiros et al. 2021, Naghi et al. 2024). The dependent variable is the monthly US inflation rate, measured as the log-difference of the Consumer Price Index (CPI) (Medeiros et al. 2021, Huang et al. 2025).

$$\pi_t = \log(P_t) - \log(P_{t-1})$$

where π_t and P_t denote the inflation rate and price index at month t, respectively, and CPI is the chosen price index in this case. Following Medeiros et al. (2021), I evaluate the inflation forecasting performance of every model in the full out-of-sample window from January 1990 to September 2025 and in two subsamples of the low-volatility inflation period from January 1990 to December 1999 and of the high-volatility inflation period from January 2000 to September 2025. These two subsamples correspond to the gray- and red-shaded segments of Figure 1, respectively. Overall, the monthly inflation rate is considerably more volatile after 2000 ($\sigma = 0.30\%$) than during the 1990s ($\sigma = 0.16\%$). This reflects the economic disruptions associated with the global financial crisis, the COVID-19 pandemic, and the Russia-Ukraine war. Table 1 provides the descriptive statistics of the US CPI inflation for the two evaluation subsamples.

4.2 Hyperparameters Tuning

The estimates and forecasts of ML models are highly sensitive to hyperparameter values that heavily influence their forecasting accuracy (Doojav and Narmandakh 2025). To address this issue, I use the approach in Huang et al. (2025) to train models and tune hyperparameters for ML methods. Table 2 below presents the search space for ML model hyperparameters. Specifically, for all shrinkage methods, the search grids of the penalty term λ are manually set, and they are relatively different from those in the literature to ensure suitability for the large training window and to avoid numerical convergence issues when training models in a rolling window scheme, where the algorithms have to fit thousands of models in each window per forecasting horizon. Therefore, the search space of Ridge ranges from 10^{-3} to 10^{4}, while Elastic Net, LASSO, and adaptive LASSO share the same penalty grid of log-spaced values. On the other hand, the tree-based models rely on the suggested hyperparameter search space in Doojav and Narmandakh (2025), with only minor variations. However, there are also other constant parameters following the idea of Medeiros et al. (2021), such as α for the Elastic Net model, which is fixed to 0.5, and the default weights of the adaptive version of LASSO. Both tree-based models use the fixed hyperparameter $n\_estimators = 500$, which is the number of trees for the case of RF and the maximum number of boosting trees for XGBoost (Huang et al. 2025, Medeiros et al. 2021). Another default parameter is $min\_samples\_leaf = 5$ such that each leaf node in RF has at least five observations (Doojav and Narmandakh 2025, Huang et al. 2025). A combination of search spaces for key hyperparameters and default values is employed in this thesis to balance computational efficiency and forecast accuracy.

Table 2: Hyperparameter Grids and Fixed Parameters for each ML model

Model	Search Space	Fixed settings
Ridge	$\lambda \in \{0.001,\ 0.01,\ 0.1,\ 1,\ 10,\ 100,\ 1000,\ 10000\}$	–
LASSO	$\lambda \in$ 12 log-spaced values in $[10^{-5},\ 1]$	–
adaLASSO	$\lambda \in$ 12 log-spaced values in $[10^{-5},\ 1]$	weights $w_i = 1/(\lvert\tilde{\beta}_{i,h}\rvert + 1/\sqrt{T})$
Elastic Net	$\lambda \in$ 12 log-spaced values in $[10^{-5},\ 1]$	$\alpha = 0.5$
Random Forest	max features $\in \{0.1,\ 0.2,\ 0.3,\ 0.5\}$	trees $= 500$; min. leaf $= 5$
XGBoost	max depth $\in \{2,\ 4,\ 6\}$; learning rate $\in \{0.1,\ 0.2,\ 0.3\}$	estimators $= 500$
adaLASSO–RF	selection $\lambda \in [10^{-5}, 1]$; max features $\in \{0.1,\ 0.2,\ 0.3,\ 0.5\}$	trees $= 500$; min. leaf $= 5$
adaLASSO–XGBoost	selection $\lambda \in [10^{-5}, 1]$; max depth $\in \{2, 4, 6\}$; learning rate $\in \{0.1, 0.2, 0.3\}$	estimators $= 500$

Following Huang et al. (2025), a rolling-window procedure is employed for modeling and hyperparameter tuning. Each model is estimated on a fixed-length training window using the last R_h months. This procedure then yields an h-month-ahead inflation forecast and rolls the window forward by one month $t + 1$, while holding the in-sample size constant, and this process repeats through the sample. As shown in Figure 2, there are two loops, where the outer loop is a series of rolling windows to train and test the model. Following chronological order, the first 80% of observations of each training window become the training subset, while the remaining 20% is the validation set for tuning ML hyperparameters. The reason for this 80% cutoff within the inner loop is that this partition provides a reasonable proportion of data for training the model while ensuring a sufficient portion for model validation, and this approach is common in macroeconomic time series forecasting (Doojav and Narmandakh 2025). I first fit each ML model to the training subset for every possible hyperparameter value in the predefined search space in Table 2 and select the optimal parameter value whose validation RMSE is the lowest. Finally, the model is refit on the entire training window using the chosen hyperparameter to produce an inflation forecast. This procedure guarantees that every model is trained on past data and tested on new and unseen data from subsequent months to assess inflation-forecasting performance with its chosen hyperparameters. I also use the `joblib` package to import the parallel function to speed up training and tuning of computationally expensive ML models, including RF, XGBoost, adaLASSO-RF, and adaLASSO-XGBoost.

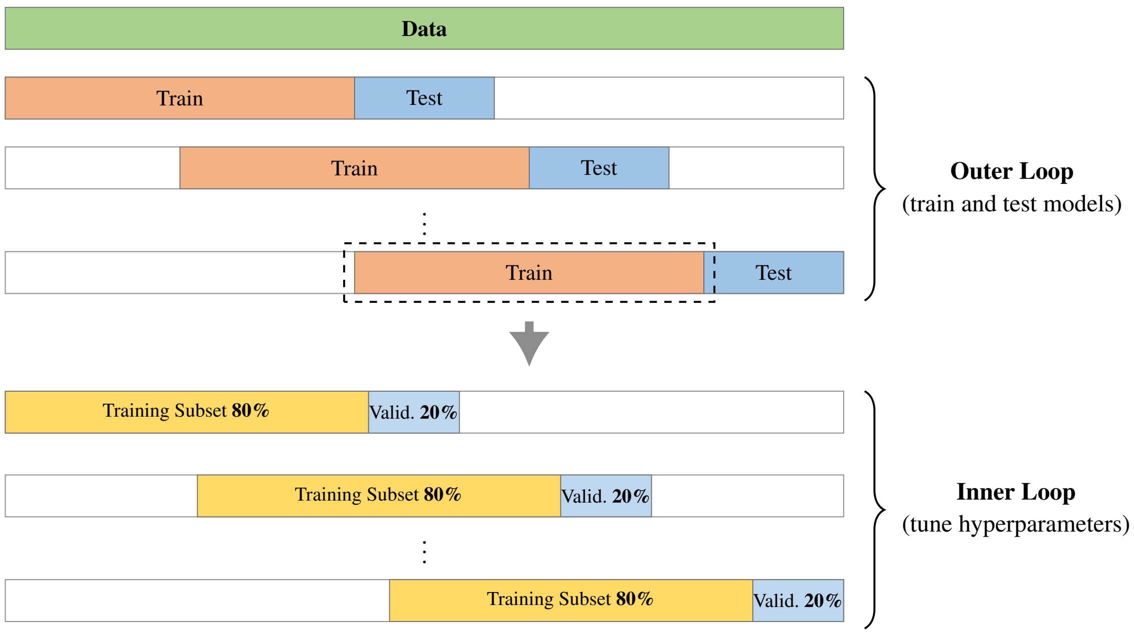


Figure 2: Rolling Window Procedure According to Huang et al. (2025)

5 Empirical Results

5.1 Evaluation Metrics

Following Medeiros et al. (2021), I evaluate how precisely every model forecasts out-of-sample using the following three metrics: the root mean squared error (RMSE), the mean absolute error (MAE), and the median absolute deviation from the median (MAD), which are defined as follows

$$\text{RMSE}_{m,h} = \sqrt{\frac{1}{T - T_0 + 1} \sum_{t=T_0}^{T} \hat{u}_{t,m,h}^2},$$

$$\text{MAE}_{m,h} = \frac{1}{T - T_0 + 1} \sum_{t=T_0}^{T} |\hat{u}_{t,m,h}|, \quad \text{and}$$

$$\text{MAD}_{m,h} = \text{median}\left[|\hat{u}_{t,m,h} - \text{median}(\hat{u}_{t,m,h})|\right].$$

where $\hat{u}_{t,m,h} = \hat{\pi}_{t,m,h} - \pi_t$ is the forecast error of model m at month t and it relies on data available through period $t - h$. The evaluation sample period runs from T_0 to T, and the divisor $T - T_0 + 1$ is the number of out-of-sample inflation forecasts over which each evaluation metric averages or takes the median. The inclusion of MAD alongside the two other metrics is because it is based on medians instead of means; therefore, MAD is insensitive to extreme forecast errors.

5.2 Interpretation

5.2.1 Full sample analysis

Table 3 reports the results of the comparison of every model in terms of the RMSE, MAE, and MAD ratio relative to the RW benchmark at every forecasting horizon over the full out-of-sample period from 1990 to 2025. The best-performing models with the lowest ratio values are highlighted in bold. In terms of RMSE ratio, it is clear that RF achieves the highest forecasting accuracy for US CPI inflation across medium- to long-term horizons, particularly at h = 3, 6, 9, and 12. However, for one-month-ahead forecasting, a majority of models are outperformed by the RW model at the short horizon h = 1; the exception is AORW, whose RMSE ratio of 0.960 is the lowest at this forecasting horizon, just slightly below RF at 0.979. All the penalized linear regression models have almost the same forecasting performance. The hybrid specifications adaLASSO-RF and adaLASSO-XGBoost slightly worsen the forecasting accuracy with higher metric ratio values, compared to the standalone RF and XGBoost models, indicating that variable selection before model estimation is not useful for increasing the quality of predictions in US CPI inflation. A similar pattern emerges in the MAE ratio panel, where RF remains superior to the competitors across most horizons, and AORW performs best at h = 1. The MAD panel presents a different perspective since nearly every model has a ratio under one at almost all horizons, so RW has the poorest performance by this measure. In particular, Elastic Net and GAS models have the most precise predictions for short- and medium-term forecasts, with Elastic Net achieving a MAD ratio of 0.927 at h = 1 and GAS 0.693 at h = 3. RF still exhibits the strongest predictive power for the remaining horizons. Another key finding across these three panels for the full out-of-sample evaluation period is that the relative performance of RW worsens as the horizon grows from h = 3 to h = 12.

Table 3: Forecasting results relative to RW: Full out-of-sample (1990–2025).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.051	0.778	0.763	0.752	0.737
AORW	**0.960**	0.767	0.757	0.763	0.747
GAS	1.022	0.763	0.758	0.752	0.744
Ridge	1.000	0.784	0.754	0.754	0.725
LASSO	1.029	0.803	0.756	0.740	0.731
adaLASSO	1.031	0.773	0.745	0.751	0.707
ElasticNet	1.048	0.781	0.772	0.755	0.713
RF	0.979	**0.727**	**0.714**	**0.708**	**0.684**
XGBoost	1.073	0.788	0.755	0.725	0.786
adaLASSO-RF	1.082	0.746	0.748	0.772	0.701
adaLASSO-XGB	1.176	0.788	0.794	0.830	0.749
Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.009	0.748	0.760	0.775	0.741
AORW	**0.939**	0.754	0.764	0.774	0.733
GAS	0.986	0.731	0.762	0.763	0.737
Ridge	0.970	0.755	0.767	0.770	0.726
LASSO	0.988	0.769	0.760	0.755	0.713
adaLASSO	0.997	0.758	0.760	0.762	0.687
ElasticNet	1.006	0.759	0.771	0.763	0.697
RF	0.954	**0.721**	**0.721**	**0.712**	**0.667**
XGBoost	1.041	0.795	0.772	0.733	0.772
adaLASSO-RF	0.995	0.732	0.765	0.790	0.694
adaLASSO-XGB	1.073	0.785	0.833	0.854	0.767
Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.945	0.738	0.719	0.784	0.661
AORW	0.931	0.776	0.736	0.742	0.653
GAS	0.965	**0.693**	0.717	0.736	0.660
Ridge	0.988	0.756	0.765	0.811	0.721
LASSO	0.961	0.785	0.733	0.771	0.647
adaLASSO	0.974	0.793	0.736	0.718	0.642
ElasticNet	**0.927**	0.776	0.745	0.749	0.649
RF	0.947	0.725	**0.699**	**0.703**	**0.637**
XGBoost	1.038	0.849	0.749	0.730	0.720
adaLASSO-RF	0.949	0.729	0.739	0.740	0.656
adaLASSO-XGB	0.946	0.779	0.855	0.780	0.725

Table 4: Forecasting results relative to RW: Less-volatile subsample (1990–1999).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.898	0.876	0.800	0.910	0.888
AORW	**0.785**	**0.800**	0.776	0.826	0.729
GAS	0.846	0.853	0.831	0.942	0.867
Ridge	0.906	0.893	0.846	0.895	0.830
LASSO	0.873	0.870	0.816	0.907	0.832
adaLASSO	0.896	0.891	0.816	0.930	0.810
ElasticNet	0.892	0.885	0.848	0.898	0.820
RF	0.859	0.832	0.783	**0.820**	**0.710**
XGBoost	0.899	0.869	0.851	0.843	0.810
adaLASSO-RF	0.822	0.807	**0.770**	0.935	0.760
adaLASSO-XGB	0.898	0.847	0.792	0.933	0.869

Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.949	0.933	0.862	1.032	1.002
AORW	**0.799**	**0.808**	**0.777**	**0.863**	**0.746**
GAS	0.882	0.907	0.925	1.083	0.984
Ridge	0.993	0.953	0.934	0.990	0.901
LASSO	0.935	0.926	0.910	1.003	0.852
adaLASSO	0.965	0.954	0.901	1.048	0.831
ElasticNet	0.968	0.939	0.947	1.013	0.833
RF	0.928	0.913	0.837	0.904	0.764
XGBoost	0.937	0.930	0.902	0.883	0.841
adaLASSO-RF	0.863	0.879	0.833	1.038	0.807
adaLASSO-XGB	0.945	0.922	0.847	0.979	0.934

Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.784	0.630	0.703	0.867	0.636
AORW	0.695	0.706	0.727	0.889	0.659
GAS	0.749	0.684	0.668	0.826	**0.593**
Ridge	0.793	0.804	0.747	0.885	0.760
LASSO	0.800	0.830	0.707	0.884	0.756
adaLASSO	0.810	0.821	0.705	0.902	0.645
ElasticNet	0.844	0.790	0.701	0.869	0.723
RF	**0.632**	**0.558**	**0.636**	0.755	0.638
XGBoost	0.754	0.811	0.733	**0.740**	0.819
adaLASSO-RF	0.686	0.699	0.645	0.843	0.676
adaLASSO-XGB	0.688	0.901	0.825	0.803	0.870

Table 5: Forecasting results relative to RW: More-volatile subsample (2000–2025).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.070	0.769	0.759	0.739	0.720
AORW	**0.981**	0.764	0.755	0.758	0.749
GAS	1.043	0.755	0.751	0.735	0.730
Ridge	1.011	0.775	0.744	0.741	0.713
LASSO	1.048	0.797	0.750	0.726	0.720
adaLASSO	1.048	0.763	0.738	0.735	0.696
ElasticNet	1.068	0.771	0.765	0.742	0.702
RF	0.994	**0.718**	**0.707**	**0.698**	**0.682**
XGBoost	1.094	0.781	0.746	0.715	0.783
adaLASSO-RF	1.112	0.741	0.746	0.758	0.695
adaLASSO-XGB	1.208	0.783	0.794	0.821	0.736
Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.023	0.714	0.740	0.728	0.686
AORW	0.972	0.745	0.762	0.757	0.730
GAS	1.011	0.698	0.730	0.703	0.685
Ridge	0.964	0.719	0.733	0.729	0.690
LASSO	1.001	0.741	0.730	0.709	0.684
adaLASSO	1.004	0.721	0.731	0.709	0.657
ElasticNet	1.015	0.726	0.735	0.716	0.668
RF	**0.960**	**0.686**	**0.698**	**0.677**	**0.646**
XGBoost	1.066	0.770	0.746	0.705	0.758
adaLASSO-RF	1.026	0.705	0.751	0.744	0.670
adaLASSO-XGB	1.103	0.760	0.830	0.831	0.732
Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.034	0.754	0.729	0.700	0.622
AORW	1.035	0.749	0.712	0.700	0.671
GAS	1.050	0.690	0.706	0.625	0.625
Ridge	1.020	**0.673**	0.755	0.712	0.680
LASSO	1.004	0.782	0.696	0.685	0.631
adaLASSO	0.982	0.751	0.689	**0.622**	**0.564**
ElasticNet	1.021	0.727	0.712	0.634	0.620
RF	**0.941**	0.746	**0.688**	0.643	0.574
XGBoost	1.033	0.832	0.704	0.667	0.672
adaLASSO-RF	0.998	0.689	0.740	0.646	0.622
adaLASSO-XGB	1.035	0.759	0.832	0.790	0.682

5.2.2 Subsample Analysis: High-Volatility vs Low-Volatility Periods

To examine whether ML models perform well during periods of high inflation volatility where inflation forecasts are difficult to obtain, subsample analyses are conducted in this section. Tables 4 and 5 present RMSE, MAE, and MAD ratios relative to the RW model during the 1990-1999 and 2000-2025 periods, respectively. The results indicate that the reported ratios of ML models are lower during the turbulent period than during the calm period at horizons beyond one month, suggesting that most models perform worse on US CPI inflation during low-volatility inflation periods relative to the RW model. ML models manage to perform well even when inflation becomes hard to predict. During the 2000-2025 period, RF dominates most of its competitors with the lowest ratios in terms of RMSE, MAE, and MAD at medium- and long-term forecasts. This strongly indicates RF is the superior specification under high-volatility inflation. In terms of the RMSE ratio, RF still confirms its dominance on medium and long horizons during high-volatility inflation periods, which exactly matches the full out-of-sample findings from the first two panels of RMSE and MAE ratio values of Table 3. Assessed by RMSE, AORW takes the lead at h = 1 and h = 3 in the calm subsample, and at h = 1 in the volatile one. During the calm periods, adaLASSO-RF achieves the highest accuracy with an RMSE ratio of 0.770 at h = 6.

Another point worth considering is that the time series model AORW outperforms other forecasting methods, with the smallest MAE ratios across all horizons during the less-volatile period of inflation. This happens because low-volatility inflation rates are easier for traditional benchmarks, which mostly rely on historical information to make inflation forecasts. By contrast, the superiority of non-linear ML models such as RF tends to yield more precise forecasts during periods of high volatility, owing to their inherent ability to handle macroeconomic uncertainties stemming from US inflation dynamics (Medeiros et al. 2021). In this case, RF demonstrates the strongest forecasting performance among its competing models, achieving the lowest MAE ratios relative to RW across all horizons. All the shrinkage models have almost the same predictive performance in both panels (a) and (b) during high- and low-inflation volatility periods because the figures for RMSE and MAE ratios are relatively similar across regularized regression methods. Moreover, other models outperform RF in terms of the MAD ratio. For example, Ridge attains the smallest MAD ratio at h = 3 (0.673), and adaptive LASSO at h = 9 and h = 12 (0.622 and 0.564) during the 2000-2025 period. XGBoost slightly outperforms RF at h = 9 in the less-volatile subsample, with a MAD ratio of 0.740 against 0.755 of RF. The same applies to the GAS model for twelve-month forecasts. Furthermore, the plain tree-based models outperform their hybrid counterparts across most panels in both subsample periods; therefore, this result closely mirrors the full-sample result in Table 3.

Model	Forecasting precision					
	Less Volatile (1990–1999)			More Volatile (2000–2025)		
	Aver RMSE	Aver MAE	Aver MAD	Aver RMSE	Aver MAE	Aver MAD
RW	1.000	1.000	1.000	1.000	1.000	1.000
AR(p)	0.874	0.956	0.724	0.811	0.778	0.768
AORW	**0.783**	**0.798**	0.735	0.801	0.793	0.774
GAS(1,1)	0.868	0.956	0.704	0.803	0.765	0.739
Ridge	0.874	0.955	0.798	0.797	0.767	0.768
LASSO	0.860	0.925	0.795	0.808	0.773	0.759
adaLASSO	0.869	0.940	0.777	0.796	0.765	0.722
Elastic Net	0.869	0.940	0.785	0.810	0.772	0.743
RF	0.801	0.869	**0.644**	**0.760**	**0.733**	**0.718**
XGBoost	0.854	0.899	0.771	0.824	0.809	0.782
adaLASSO-RF	0.819	0.884	0.709	0.810	0.779	0.739
adaLASSO-XGB	0.868	0.925	0.818	0.869	0.851	0.820

Table 6: Average RMSE, MAE, and MAD ratios relative to RW, obtained by computing the mean of ratios across all five forecasting horizons, under the less- and more-volatile subsample periods. The average ratios for the best models are highlighted in bold.

Table 6 reports the average RMSE, MAE, and MAD ratios relative to RW across all five horizons for each model in less- and more-volatile subsample periods. These average ratios are consistent with the results in Tables 4 and 5. RF is the best-performing model, as indicated by the smallest average ratios on all three evaluation metrics during the more-volatile period and the lowest average MAD during the less-volatile period. The dominance of this model over all classical time-series benchmarks and all other ML models in the volatile subsample is consistent with the findings of Medeiros et al. (2021), who confirm the superiority of RF for US inflation over the 1990-2015 full out-of-sample period and during the more volatile 2001-2015 period. However, ML models are not always the best forecasting methods. The benchmark AORW is the winning method that attains the lowest average RMSE and MAE in the low-volatility subsample and outperforms every ML specification. Furthermore, both hybrid models and XGBoost perform no better than the best benchmark on all three metrics in both subsamples. In terms of RMSE and MAE, most models have lower average ratios in the fluctuating period than in the calm period, implying that accuracy gains relative to RW become smaller when inflation is less challenging to forecast. Finally, the hybrid ML specifications adaLASSO-RF and adaLASSO-XGBoost underperform relative to standalone RF and XGBoost in every comparison, strongly suggesting that variable selection does not improve the accuracy of ML models for inflation forecasting.

	Forecasting horizon				
Benchmark	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$
Panel (a): Squared loss					
RW	-0.29	-3.59^{***}	-2.72^{***}	-3.10^{***}	-3.72^{***}
AR(p)	-3.06^{***}	-1.87^{*}	-1.80^{*}	-1.86^{*}	-2.62^{***}
AORW	0.91	-1.48	-1.73^{*}	-2.15^{**}	-2.07^{**}
GAS(1,1)	-1.86^{*}	-1.66^{*}	-1.89^{*}	-2.84^{***}	-3.71^{***}
Panel (b): Absolute loss					
RW	-0.96	-5.36^{***}	-4.35^{***}	-4.05^{***}	-5.25^{***}
AR(p)	-2.80^{***}	-1.64	-1.75^{*}	-2.93^{***}	-3.37^{***}
AORW	0.64	-1.46	-1.57	-1.70^{*}	-1.83^{*}
GAS(1,1)	-1.63	-0.62	-2.15^{**}	-3.01^{***}	-3.52^{***}

Table 7: Forecasting result: Diebold–Mariano test results over the full out-of-sample period (1990–2025). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 shows the DM test statistics results when comparing RF against each time-series model at every forecasting horizon over the full out-of-sample period from 1990 to 2025. The loss differential is defined as the loss of RF minus the loss of each benchmark. Overall, the DM test statistics are negative in nearly all cells, except at the one-month-ahead forecasting horizon for AORW, indicating that AORW is the only time-series model to outperform the RF model, but only insignificantly. In particular, RF has significantly better forecasting performance than RW at horizon h = 3 and onward under both loss functions. However, these two models are statistically indistinguishable at horizon h = 1, and this matches the RMSE ratio of 0.979 for RF in Table 3, which is barely below one. RF significantly outperforms the AR benchmark, particularly at a 1% significance level at h = 1 and h = 12 under both loss functions. Under squared loss, the dominance of RF over AR is significant at 10% for h = 3-9; however, under absolute loss, h = 6 and h = 9 are significant at 10% and 1% respectively, while it is insignificant at h = 3 (-1.64). Turning attention to AORW, it produces forecasts that are insignificantly more accurate than RF at h = 1. However, RF outperforms AORW as the horizon increases, especially for h = 9 and 12 at the 5% significance level under the squared loss function and 10% under the absolute loss function. Finally, there is statistically strong evidence that RF beats the GAS model in inflation forecasting accuracy for long-term horizons, with 9-month and 12-month forecasts at the 1% significance level. These findings confirm that RF is statistically significantly superior to all four benchmark models at nine- and twelve-month forecasts. AORW remains the best forecasting method for short forecasting horizons, aligning with the results from the ratio and average RMSE, MAE, and MAD tables.

5.3 Variable Importance

A variable importance analysis is further conducted to identify the most informative variables for predicting US CPI inflation. Based on the results in Section 5.2, I examine the predictors selected by the following four best-performing models: RF, XGBoost, RR, and adaLASSO. In particular, RR is a non-sparse method that only shrinks the regression coefficients (Hoerl and Kennard 1970) to nearly zero; in contrast, adaLASSO induces sparsity that sets coefficients of the least contributive variables to exactly zero (Medeiros and Mendes 2016). Both of these penalized regression models are linear specifications, whereas RF and XGBoost are nonlinear. This set of ML methods closely follows Medeiros et al. (2021), who used adaLASSO, RF, and RR to measure variable importance. Furthermore, XGBoost, RR, and RF are considered in the Mongolian inflation forecasting study of Doojav and Narmandakh (2025). In addition, RF and RR are considered in the paper by Huang et al. (2025) to determine key variables for predicting China's CPI and PPI inflation.

The variable importance procedure has two stages. In the first stage, for shrinkage-based models (RR and adaLASSO), variable importance is evaluated by the absolute magnitude of the estimated coefficients because all features are standardized before estimation; therefore, these coefficient magnitudes are comparable (Huang et al. 2025). For XGBoost, the feature importance is measured by gain, which represents the loss reduction by a given feature, accumulated over all splits related to that feature, as proposed by Chen and Guestrin (2016). I employ impurity-based feature importance analysis for each variable in the RF model, following Breiman (2001). This method measures the contribution of each feature to reducing impurity.

The second stage is to measure and compare the importance of variable groups. I use the following aggregation steps, as described in Huang et al. (2025). First, I add four lags of a variable together to obtain a single importance score for that variable. Second, I assign each resulting variable to the following eight variable groups. According to McCracken and Ng (2016), these are housing, stock market, interest and exchange rates, labor market, output and income, consumption and orders, money and credit, and prices. US CPI inflation lags are treated as a separate autoregressive group. Third, I compute the within-group mean for each of the above-mentioned groups to measure variable group importance (Huang et al. 2025). The tables and figures below report group importance for RR, adaLASSO, RF, and XGBoost at each forecasting horizon for the full sample and two subsamples. They also show the corresponding average importance across all horizons and their ranks for US CPI inflation forecasts.

5.3.1 Interpretation

In general, the four most important variable groups vary across models, evaluation periods, and horizons, although several groups reappear. RF delivers the most stable feature importance among the four models. The AR group is the most important and dominant predictor group, as it consistently shows up among the top four groups in almost all panels, ranking number one in eight of the 12 table panels, implying that inflation lags are the most reliable explanatory factor and that US CPI inflation is strongly persistent. In contrast, the price group has the least predictive power, which is barely in the top four and usually has the lowest rankings across the result panels, except in the case of low-inflation-volatility periods for adaptive LASSO from 1990 to 1999 in Figure 6b, where it ranks third, just above the labor market.

Another notable point is that the variable group importance result is similar for the two tree-based models. In particular, the top four are identical across the full sample (AR term, housing, labor market, interest and exchange rates). During the 1990-1999 period, when inflation is less volatile, the shared groups are AR, housing, and money and credit. Furthermore, housing is consistently ranked second under the full sample and the less-volatile subsample for both RF and XGBoost. The interest and exchange rate group of predictors is also common in both the full out-of-sample period and the high-inflation-volatility period for both the tree-based models, as shown in Tables 10 and 11, and the bar and line charts of Figures 3 and 4.

On the other hand, the dominance of money and credit occurs in both shrinkage models. Specifically, this predictor group is ranked first in the full and high-volatility panels when using Ridge Regression and ranked first in the case of adaptive LASSO during the period of high-volatility inflation, as shown in Tables 8 and 9 and Figures 5 and 6. This contrasts with tree-based models, which rarely include money and credit in the top four, whereas RF and XGBoost instead place a noticeable emphasis on AR terms and housing. In the high-volatility subsample, Ridge removes the AR group from its top four and ranks it seventh with an average importance of 8.9%, slightly above prices and housing, as illustrated by the right line chart of Figure 5b. Another observation arises from the middle panel of adaLASSO in the less-volatile inflation periods of the line chart in Figure 6b, where output and income peak at 67.5% as the highest figure across all panels at the long-term forecasting horizon h = 9 and are the top predictor group with a mean importance of 20.3% across all horizons and therefore outperform the inflation lags. Finally, adaptive LASSO is the model that most often sets whole groups of predictors to zero importance. At the long horizon h = 9 in the calm subsample, the AR terms, labor market, housing, consumption and orders, money and credit, interest and exchange rates, and stock market groups receive zero importance, as shown by Table 9 and the middle panels of Figures 6a and 6b. This demonstrates the sparsity of adaLASSO, which can retain a reduced set of highly relevant predictors and eliminate the rest (Medeiros and Mendes 2016).

Group	Forecasting horizon $h=1$	$h=3$	$h=6$	$h=9$	$h=12$	Avg	Rank
Panel (a): Full out-of-sample 1990–2025							
AR (inflation lags)	5.9%	6.3%	19.5%	15.0%	11.6%	**11.6%**	**4**
Output & income	12.7%	13.0%	9.9%	8.8%	7.3%	10.4%	6
Labor market	13.8%	14.6%	11.9%	11.4%	14.1%	**13.1%**	**2**
Housing	8.1%	10.6%	5.5%	6.3%	7.0%	7.5%	9
Consumption & orders	13.8%	9.5%	9.3%	10.2%	9.8%	10.5%	5
Money & credit	13.1%	17.4%	15.2%	14.8%	16.2%	**15.4%**	**1**
Interest & exch. rates	11.5%	11.3%	8.0%	7.7%	10.3%	9.8%	7
Prices	8.7%	7.1%	6.7%	10.5%	10.3%	8.7%	8
Stock market	12.4%	10.1%	13.9%	15.2%	13.4%	**13.0%**	**3**
Panel (b): High volatility 2000–2025							
AR (inflation lags)	2.5%	4.6%	11.6%	12.7%	13.3%	8.9%	7
Output & income	13.0%	14.2%	11.0%	9.4%	7.8%	11.1%	5
Labor market	13.8%	14.0%	12.4%	10.8%	13.6%	**12.9%**	**3**
Housing	9.4%	11.7%	6.1%	6.6%	8.2%	8.4%	9
Consumption & orders	13.6%	9.7%	9.9%	11.1%	8.4%	10.5%	6
Money & credit	13.5%	17.4%	15.5%	15.3%	15.4%	**15.4%**	**1**
Interest & exch. rates	12.7%	13.3%	10.0%	9.6%	10.6%	**11.2%**	**4**
Prices	8.1%	6.8%	7.3%	10.0%	10.3%	8.5%	8
Stock market	13.5%	8.3%	16.1%	14.7%	12.4%	**13.0%**	**2**
Panel (c): Low volatility 1990–1999							
AR (inflation lags)	18.8%	14.7%	22.1%	20.7%	15.2%	**18.3%**	**1**
Output & income	9.3%	8.3%	10.1%	10.9%	9.9%	9.7%	6
Labor market	12.4%	13.7%	11.5%	10.3%	10.4%	**11.7%**	**3**
Housing	10.1%	10.8%	11.1%	4.9%	5.6%	8.5%	9
Consumption & orders	8.4%	9.0%	9.6%	11.2%	12.7%	**10.2%**	**4**
Money & credit	9.9%	13.1%	10.4%	12.9%	14.5%	**12.2%**	**2**
Interest & exch. rates	12.1%	12.2%	9.7%	8.0%	6.3%	9.7%	8
Prices	8.5%	8.6%	7.4%	11.2%	12.8%	9.7%	7
Stock market	10.6%	9.5%	8.0%	10.0%	12.5%	10.1%	5

Table 8: Variable importance for US CPI inflation based on the Ridge regression. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

Group	$h=1$	$h=3$	$h=6$	$h=9$	$h=12$	Avg	Rank
	Forecasting horizon						
Panel (a): Full out-of-sample 1990–2025							
AR (inflation lags)	0.0%	0.0%	64.1%	19.1%	55.5%	**27.7%**	**1**
Output & income	15.3%	25.0%	2.5%	6.9%	0.0%	9.9%	5
Labor market	18.3%	19.7%	4.0%	7.8%	10.1%	**12.0%**	**3**
Housing	3.1%	5.9%	2.8%	2.6%	6.8%	4.2%	8
Consumption & orders	24.2%	17.2%	2.6%	9.5%	5.4%	**11.8%**	**4**
Money & credit	5.7%	25.5%	13.5%	26.3%	17.7%	**17.7%**	**2**
Interest & exch. rates	10.2%	4.8%	0.5%	3.9%	4.4%	4.8%	7
Prices	1.1%	1.9%	0.5%	7.8%	0.1%	2.3%	9
Stock market	22.1%	0.0%	9.6%	15.9%	0.0%	9.5%	6
Panel (b): High volatility 2000–2025							
AR (inflation lags)	0.0%	0.0%	37.8%	0.0%	23.1%	**12.2%**	**2**
Output & income	17.7%	24.9%	6.0%	0.0%	3.5%	**10.4%**	**4**
Labor market	15.9%	12.8%	5.3%	7.5%	13.2%	**10.9%**	**3**
Housing	2.3%	7.9%	4.0%	28.5%	8.5%	10.2%	6
Consumption & orders	8.0%	8.0%	6.3%	10.1%	2.2%	6.9%	8
Money & credit	15.2%	28.4%	17.0%	33.9%	31.6%	**25.2%**	**1**
Interest & exch. rates	10.8%	11.5%	2.1%	12.1%	12.4%	9.8%	7
Prices	3.2%	2.6%	1.6%	7.8%	4.7%	4.0%	9
Stock market	27.0%	3.9%	20.0%	0.0%	0.8%	10.3%	5
Panel (c): Low volatility 1990–1999							
AR (inflation lags)	51.5%	0.0%	49.1%	0.0%	0.0%	**20.1%**	**2**
Output & income	6.7%	8.3%	5.7%	67.5%	13.5%	**20.3%**	**1**
Labor market	12.3%	24.1%	8.4%	0.0%	11.6%	**11.3%**	**4**
Housing	1.1%	2.9%	9.6%	0.0%	0.8%	2.9%	9
Consumption & orders	1.7%	5.6%	11.8%	0.0%	17.4%	7.3%	8
Money & credit	4.1%	7.2%	4.9%	0.0%	23.0%	7.9%	6
Interest & exch. rates	10.4%	18.8%	7.7%	0.0%	2.3%	7.8%	7
Prices	3.1%	10.4%	1.7%	32.5%	15.7%	**12.7%**	**3**
Stock market	9.2%	22.6%	1.1%	0.0%	15.7%	9.7%	5

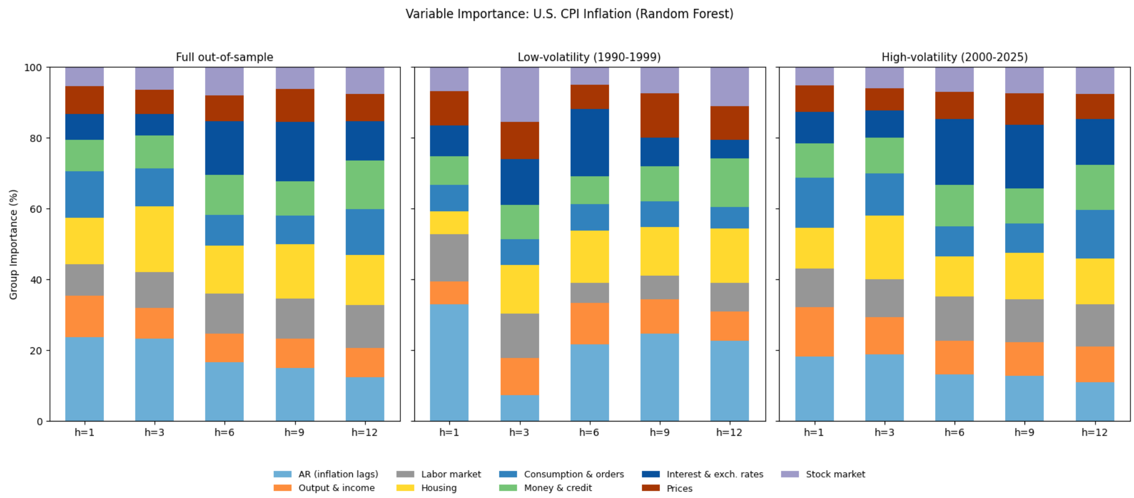
Table 9: Variable importance for US CPI inflation based on the adaptive LASSO. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

Group	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	Avg	Rank
	Forecasting horizon						
Panel (a): Full out-of-sample 1990–2025							
AR (inflation lags)	23.5%	23.2%	16.5%	14.9%	12.2%	**18.0%**	**1**
Output & income	11.7%	8.7%	8.0%	8.2%	8.2%	9.0%	7
Labor market	8.9%	10.1%	11.3%	11.3%	12.1%	**10.7%**	**4**
Housing	13.3%	18.6%	13.7%	15.5%	14.3%	**15.1%**	**2**
Consumption & orders	13.0%	10.5%	8.5%	8.1%	13.0%	10.6%	6
Money & credit	9.0%	9.3%	11.5%	9.7%	13.6%	10.6%	5
Interest & exch. rates	7.2%	6.2%	15.0%	16.8%	11.1%	**11.3%**	**3**
Prices	7.8%	7.0%	7.3%	9.1%	7.8%	7.8%	8
Stock market	5.6%	6.5%	8.2%	6.4%	7.7%	6.9%	9
Panel (b): High volatility 2000–2025							
AR (inflation lags)	18.1%	18.8%	13.0%	12.7%	10.8%	**14.7%**	**1**
Output & income	13.9%	10.4%	9.6%	9.5%	10.0%	10.7%	7
Labor market	11.0%	10.6%	12.4%	12.0%	12.0%	**11.6%**	**4**
Housing	11.4%	18.0%	11.3%	13.1%	12.8%	**13.3%**	**2**
Consumption & orders	14.2%	12.1%	8.6%	8.4%	13.9%	11.4%	5
Money & credit	9.7%	10.1%	11.6%	9.7%	12.7%	10.8%	6
Interest & exch. rates	8.9%	7.7%	18.6%	18.1%	13.0%	**13.2%**	**3**
Prices	7.5%	6.3%	7.6%	9.0%	7.0%	7.5%	8
Stock market	5.3%	6.1%	7.3%	7.5%	7.8%	6.8%	9
Panel (c): Low volatility 1990–1999							
AR (inflation lags)	32.8%	7.2%	21.6%	24.5%	22.5%	**21.7%**	**1**
Output & income	6.6%	10.4%	11.6%	9.8%	8.3%	9.3%	6
Labor market	13.3%	12.6%	5.8%	6.5%	8.0%	9.3%	8
Housing	6.5%	13.7%	14.6%	13.8%	15.5%	**12.8%**	**2**
Consumption & orders	7.4%	7.3%	7.4%	7.3%	6.0%	7.1%	9
Money & credit	8.1%	9.6%	8.0%	9.9%	13.8%	**9.9%**	**4**
Interest & exch. rates	8.7%	13.0%	19.0%	8.0%	5.2%	**10.8%**	**3**
Prices	9.6%	10.5%	7.0%	12.4%	9.5%	9.8%	5
Stock market	7.0%	15.6%	5.1%	7.6%	11.2%	9.3%	7

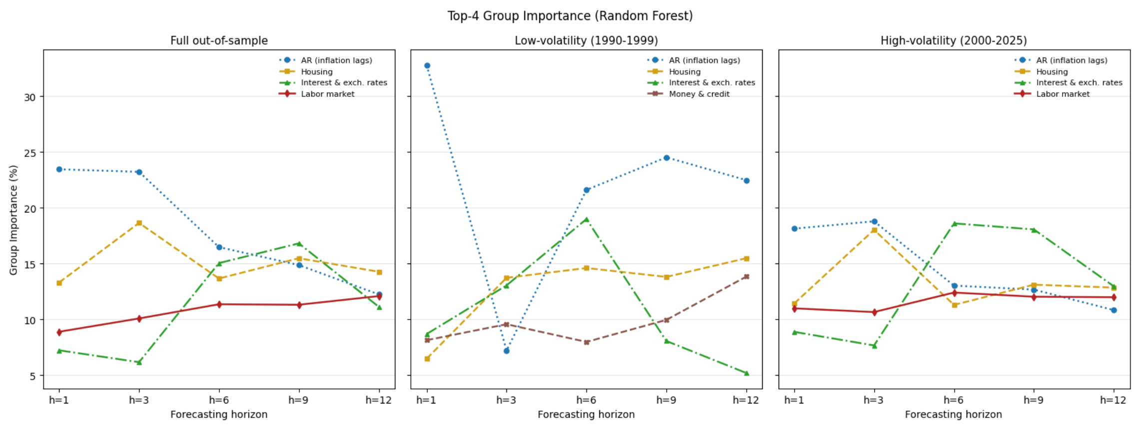
Table 10: Variable importance for US CPI inflation based on the Random Forest. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

	Forecasting horizon						
Group	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$	Avg	Rank
Panel (a): Full out-of-sample 1990–2025							
AR (inflation lags)	31.2%	17.6%	23.1%	23.1%	17.6%	**22.5%**	**1**
Output & income	9.7%	9.7%	12.0%	9.3%	7.4%	9.6%	6
Labor market	8.4%	12.0%	12.2%	8.9%	14.5%	**11.2%**	**3**
Housing	9.8%	17.4%	12.6%	14.1%	14.1%	**13.6%**	**2**
Consumption & orders	11.8%	8.7%	6.3%	7.8%	12.7%	9.5%	7
Money & credit	8.6%	9.6%	10.6%	9.3%	10.6%	9.7%	5
Interest & exch. rates	9.9%	11.3%	12.5%	8.4%	8.3%	**10.1%**	**4**
Prices	5.9%	5.6%	4.7%	8.3%	6.3%	6.2%	9
Stock market	4.8%	8.0%	6.1%	10.9%	8.5%	7.6%	8
Panel (b): High volatility 2000–2025							
AR (inflation lags)	7.3%	12.1%	13.7%	11.7%	25.7%	**14.1%**	**1**
Output & income	15.8%	10.1%	11.8%	12.2%	9.3%	**11.8%**	**3**
Labor market	11.9%	13.8%	12.7%	9.0%	8.3%	11.1%	5
Housing	7.7%	9.3%	9.4%	15.3%	12.3%	10.8%	6
Consumption & orders	13.9%	10.7%	8.6%	10.5%	12.3%	**11.2%**	**4**
Money & credit	11.4%	13.0%	7.1%	14.4%	7.7%	10.8%	7
Interest & exch. rates	12.9%	16.3%	13.6%	13.6%	10.5%	**13.4%**	**2**
Prices	8.4%	6.4%	5.6%	9.8%	4.0%	6.8%	9
Stock market	10.6%	8.2%	17.5%	3.4%	9.9%	9.9%	8
Panel (c): Low volatility 1990–1999							
AR (inflation lags)	24.4%	12.9%	28.9%	22.3%	0.0%	**17.7%**	**1**
Output & income	7.1%	8.4%	9.1%	9.5%	12.0%	9.2%	6
Labor market	9.8%	11.5%	7.0%	7.4%	9.3%	9.0%	7
Housing	8.8%	13.3%	15.8%	24.8%	23.4%	**17.2%**	**2**
Consumption & orders	16.9%	12.5%	13.6%	7.6%	8.8%	**11.9%**	**4**
Money & credit	7.9%	13.5%	8.0%	10.0%	23.2%	**12.5%**	**3**
Interest & exch. rates	5.0%	11.4%	10.2%	6.1%	7.7%	8.1%	8
Prices	10.2%	12.1%	5.6%	7.4%	15.6%	10.2%	5
Stock market	9.9%	4.3%	1.8%	4.9%	0.0%	4.2%	9

Table 11: Variable importance for US CPI inflation based on XGBoost. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

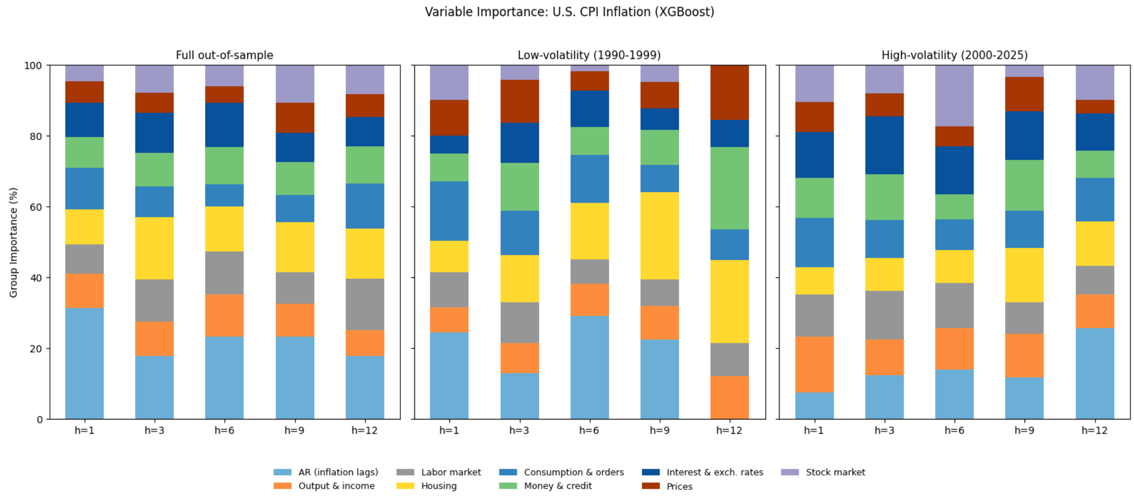


(a) The stacked bar chart shows variable group importance at each forecasting horizon for the RF model under the full sample and two subsample periods

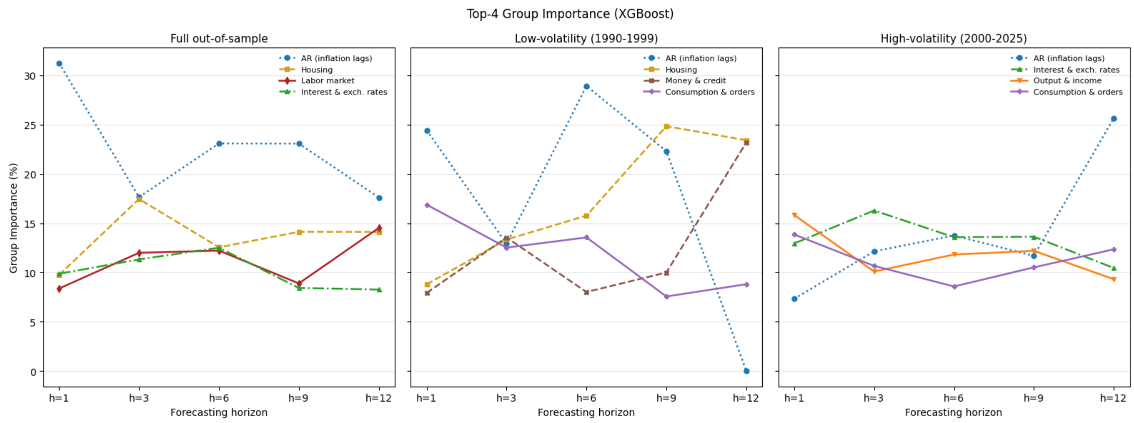


(b) The line chart shows the four most important groups over forecasting horizons for the RF model under the full sample and two subsample periods

Figure 3: Variable importance for US CPI inflation based on the Random Forest model across the full out-of-sample period and the low- and high-volatility subsample periods

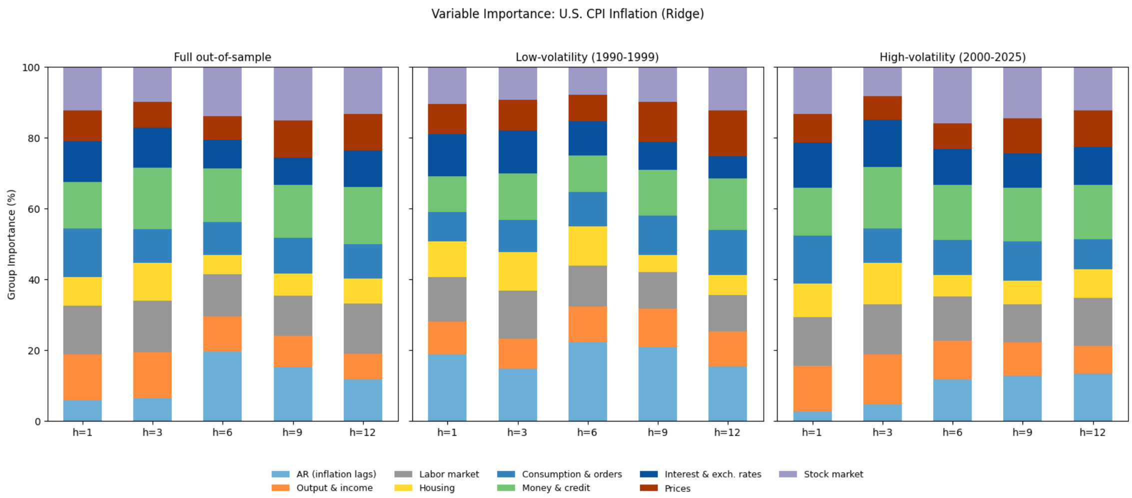


(a) The stacked bar chart shows variable group importance at each forecasting horizon for XGBoost under the full sample and two subsample periods

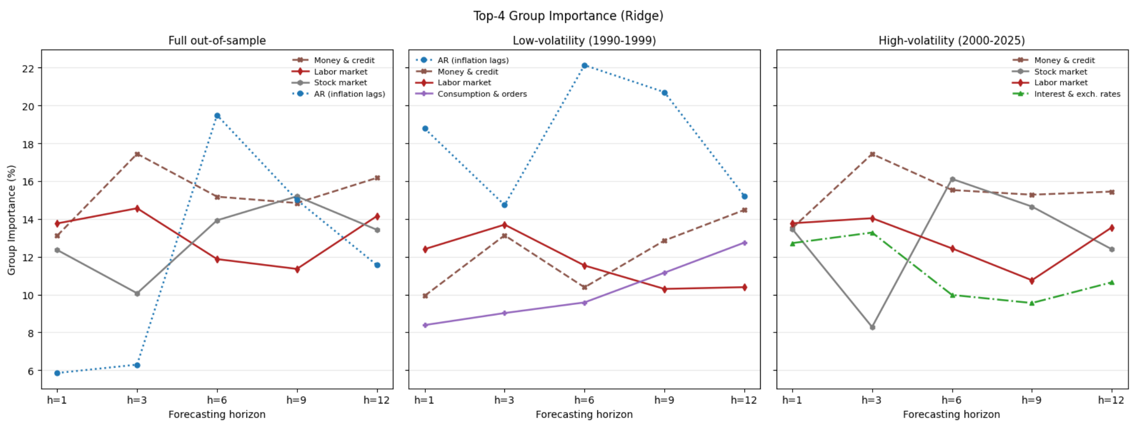


(b) The line chart shows the four most important groups over forecasting horizons for XGBoost under the full sample and two subsample periods

Figure 4: Variable importance for US CPI inflation based on the XGBoost across the full out-of-sample period and the low- and high-volatility subsample periods

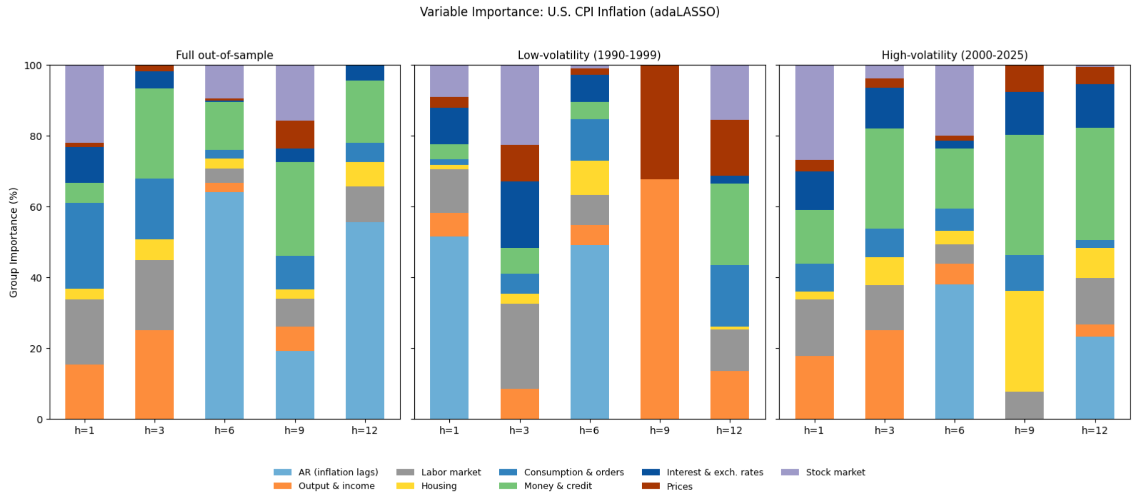


(a) The stacked bar chart shows variable group importance at each forecasting horizon for Ridge under the full sample and two subsample periods

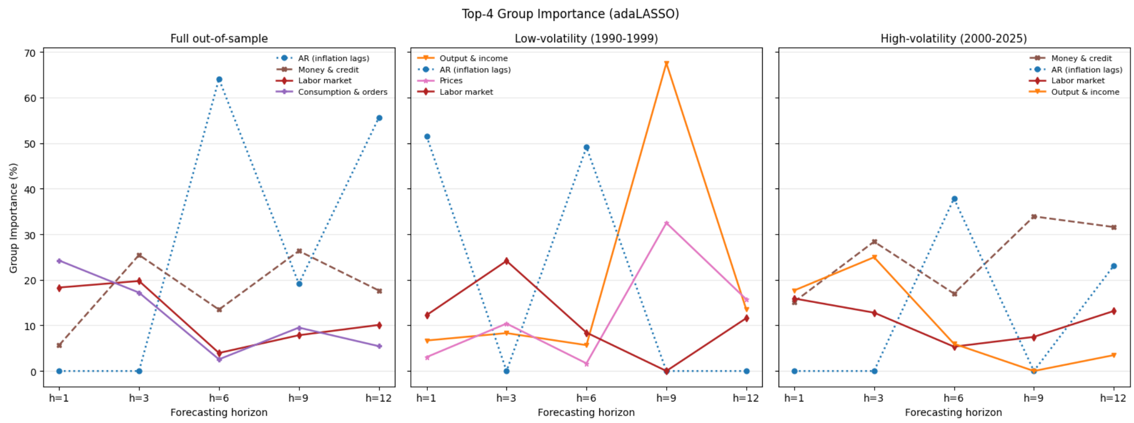


(b) The line chart shows the four most important groups over forecasting horizons for Ridge under the full sample and two subsample periods

Figure 5: Variable importance for US CPI inflation based on the Ridge Regression across the full out-of-sample period and the low- and high-volatility subsample periods



(a) The stacked bar chart shows variable group importance at each forecasting horizon for adaLASSO under the full sample and two subsample periods



(b) The line chart shows the four most important groups over the forecasting horizons for adaLASSO under the full sample and two subsample periods

Figure 6: Variable importance for US CPI inflation based on the adaptive LASSO across the full out-of-sample period and the low- and high-volatility subsample periods

6 Conclusion

This thesis demonstrates the benefits of Machine Learning (ML) models with rich data in improving inflation forecasting accuracy and identifying key predictors for US CPI inflation. The empirical analysis yields three main findings. First, when estimated on a large monthly US macroeconomic dataset, ML methods such as adaptive LASSO, Ridge, and Random Forest outperform the traditional time-series benchmarks, particularly at medium and long horizons during high-volatility inflation periods, where precise forecasts are harder to achieve. The tree-based model with superior predictive performance is the Random Forest, as it has the lowest RMSE and MAE at every horizon beyond one month during the fluctuating inflation periods. This can be explained by the natural variable selection mechanism and by its capability to handle nonlinear relationships between the predictors and the target variable. However, AORW is the special case in which the benchmark model performs better than the ML models on short- to medium-horizon forecasts, especially during less-volatile periods of inflation. These findings hold even when the Diebold-Mariano (DM) test compares the equality of inflation-forecasting accuracy between Random Forest and each benchmark model.

Second, the variable-importance analysis reveals that there is stability in the predictor groups determined by RF across horizons, in both the full sample and the subsamples. AR terms for the inflation lags are the best predictors. Apart from that, variables from the following groups, namely, money and credit, housing, interest and exchange rates, and the labor market, are the most important predictors contributing to US CPI inflation forecasts. As a result, these variables are identified as key drivers of US inflation dynamics. Finally, the standalone tree-based models beat their hybrid counterparts because adaLASSO-RF and adaLASSO-XGBoost consistently exhibit larger errors than plain RF and XGBoost. Selecting features before model estimation therefore does not improve ML inflation forecasting accuracy. Taken together, these results provide valuable insights for ML modeling and forecasting inflation in a high-dimensional macroeconomic setting, supporting policymakers and the government in controlling inflation rates to stabilize the economy and ensure sustainable growth.

Given a limited set of models, a single penalized regression used for feature selection in the first-step procedure of hybrid ML methods, and potentially poor generalization to other countries for inflation forecasting, future research should close these gaps by adopting the following initiatives. First, I could examine forecast combinations, including the simple mean, trimmed mean, and median of the individual forecasts (Bates and Granger 1969). These methods combine the strengths of each model and attenuate the impact of extreme forecasts. Therefore, the future goal is to assess whether the combined forecasting models beat the best single model. Second, a further extension is to expand the model set to include neural networks such as Multi-Layer Perceptron (MLP) (Bishop 1995) and recurrent neural networks (RNN) (Elman 1990), which are designed for high-dimensional data input and nonlinear dynamics. Third, the selection step of the hybrid design could be diversified by using LASSO or Elastic Net rather than adaLASSO to account for the influence of selecting variables before estimation in enhancing the forecasting accuracy of ML models. Finally, future research could consider assessing the real-time forecasting performance of benchmark and ML models apart from the vintage FRED-MD data, following Garcia et al. (2017) and Groen et al. (2013).

A Appendix

The following tables show the 119 series retained in the US CPI inflation forecasting study with their corresponding stationary transformations, where x_t represents a raw series and Δ denotes the first difference. These tables are classified into eight groups, following McCracken and Ng (2016).

Table 12: Output and income (16 series)

Series	Transformation	Description
RPI	$\Delta \log x_t$	Real personal income
W875RX1	$\Delta \log x_t$	Real personal income, excluding transfer receipts
INDPRO	$\Delta \log x_t$	Industrial production, total index
IPFPNSS	$\Delta \log x_t$	Industrial production, final products and nonindustrial supplies
IPFINAL	$\Delta \log x_t$	Industrial production, final products
IPCONGD	$\Delta \log x_t$	Industrial production, consumer goods
IPDCONGD	$\Delta \log x_t$	Industrial production, durable consumer goods
IPNCONGD	$\Delta \log x_t$	Industrial production, nondurable consumer goods
IPBUSEQ	$\Delta \log x_t$	Industrial production, business equipment
IPMAT	$\Delta \log x_t$	Industrial production, materials
IPDMAT	$\Delta \log x_t$	Industrial production, durable materials
IPNMAT	$\Delta \log x_t$	Industrial production, nondurable materials
IPMANSICS	$\Delta \log x_t$	Industrial production, manufacturing sector
IPB51222S	$\Delta \log x_t$	Industrial production, residential utilities
IPFUELS	$\Delta \log x_t$	Industrial production, fuels
CUMFNS	Δx_t	Capacity utilization rate in manufacturing

Table 13: Labor market (31 series)

Series	Transformation	Description
HWI	Δx_t	Help-wanted advertising index for the USA
HWIURATIO	Δx_t	Help-wanted ads relative to the number of unemployed
CLF16OV	$\Delta \log x_t$	Civilian labor force, persons aged 16 and over
CE16OV	$\Delta \log x_t$	Civilian employment, persons aged 16 and over
UNRATE	Δx_t	Civilian unemployment rate
UEMPMEAN	Δx_t	Mean duration of unemployment in weeks
UEMPLT5	$\Delta \log x_t$	The number of unemployed for under 5 weeks
UEMP5TO14	$\Delta \log x_t$	The number of unemployed for 5 to 14 weeks
UEMP15OV	$\Delta \log x_t$	The number of unemployed for 15 weeks or longer
UEMP15T26	$\Delta \log x_t$	The number of unemployed for 15 to 26 weeks
UEMP27OV	$\Delta \log x_t$	The number of unemployed for 27 weeks or longer

Series	Transformation	Description
CLAIMSx	$\Delta \log x_t$	Initial claims for unemployment insurance
PAYEMS	$\Delta \log x_t$	Total Nonfarm payroll employment
USGOOD	$\Delta \log x_t$	Payroll employment, goods-producing sector
CES1021000001	$\Delta \log x_t$	Payroll employment, mining sector
USCONS	$\Delta \log x_t$	Payroll employment, construction sector
MANEMP	$\Delta \log x_t$	Payroll employment, manufacturing sector
DMANEMP	$\Delta \log x_t$	Payroll employment, durable goods manufacturing sector
NDMANEMP	$\Delta \log x_t$	Payroll employment, nondurable goods manufacturing sector
SRVPRD	$\Delta \log x_t$	Payroll employment, service-providing sector
USTPU	$\Delta \log x_t$	Payroll employment, trade, transportation, and utilities
USWTRADE	$\Delta \log x_t$	Payroll employment, wholesale trade
USTRADE	$\Delta \log x_t$	Payroll employment, retail trade
USFIRE	$\Delta \log x_t$	Payroll employment, financial activities
USGOVT	$\Delta \log x_t$	Payroll employment, government
CES0600000007	x_t	Weekly work hours, goods-producing sector
AWOTMAN	Δx_t	Weekly overtime work hours, manufacturing sector
AWHMAN	x_t	Weekly work hours, manufacturing sector
CES0600000008	$\Delta^2 \log x_t$	Hourly earnings, goods-producing sector
CES2000000008	$\Delta^2 \log x_t$	Hourly earnings, construction sector
CES3000000008	$\Delta^2 \log x_t$	Hourly earnings, manufacturing sector

Table 14: Housing (10 series)

Series	Transformation	Description
HOUST	$\log x_t$	Housing starts, national total
HOUSTNE	$\log x_t$	Housing starts, Northeast region
HOUSTMW	$\log x_t$	Housing starts, Midwest region
HOUSTS	$\log x_t$	Housing starts, South region
HOUSTW	$\log x_t$	Housing starts, West region
PERMIT	$\log x_t$	Residential building permits, national total
PERMITNE	$\log x_t$	Residential building permits, Northeast region
PERMITMW	$\log x_t$	Residential building permits, Midwest region
PERMITS	$\log x_t$	Residential building permits, South region
PERMITW	$\log x_t$	Residential building permits, West region

Table 15: Consumption and orders (7 series)

Series	Transformation	Description
DPCERA3M086SBEA	$\Delta \log x_t$	Real personal consumption expenditures
CMRMTSPLx	$\Delta \log x_t$	Real manufacturing and trade industry sales
RETAILx	$\Delta \log x_t$	Retail and food services sales
AMDMNOx	$\Delta \log x_t$	New orders for durable goods
AMDMUOx	$\Delta \log x_t$	Unfilled orders for durable goods
BUSINVx	$\Delta \log x_t$	Total business inventories
ISRATIOx	Δx_t	Ratio of business inventories to sales

Table 16: Money and credit (13 series)

Series	Transformation	Description
M1SL	$\Delta^2 \log x_t$	M1 money stock
M2SL	$\Delta^2 \log x_t$	M2 money stock
M2REAL	$\Delta \log x_t$	M2 money stock in real terms
BOGMBASE	$\Delta^2 \log x_t$	Monetary base
TOTRESNS	$\Delta^2 \log x_t$	Total reserves held by depository institutions
NONBORRES	$\Delta(x_t/x_{t-1} - 1)$	Nonborrowed reserves of depository institutions
BUSLOANS	$\Delta^2 \log x_t$	Commercial and industrial loans
REALLN	$\Delta^2 \log x_t$	Real estate loans at commercial banks
NONREVSL	$\Delta^2 \log x_t$	Nonrevolving consumer credit outstanding
CONSPI	Δx_t	Nonrevolving consumer credit relative to personal income
DTCOLNVHFNM	$\Delta^2 \log x_t$	Outstanding motor vehicle loans to consumers
DTCTHFNM	$\Delta^2 \log x_t$	Outstanding consumer loans and leases in total
INVEST	$\Delta^2 \log x_t$	Securities held in bank credit for all commercial banks

Table 17: Interest and exchange rates (19 series)

Series	Transformation	Description
FEDFUNDS	Δx_t	Effective federal funds rate
TB3MS	Δx_t	3-month treasury bill rate
TB6MS	Δx_t	6-month treasury bill rate
GS1	Δx_t	1-year treasury constant maturity rate
GS5	Δx_t	5-year treasury constant maturity rate
GS10	Δx_t	10-year treasury constant maturity rate
AAA	Δx_t	Corporate bond yield, Moody's Aaa rating
BAA	Δx_t	Corporate bond yield, Moody's Baa rating

Series	Transformation	Description
TB3SMFFM	x_t	Spread: 3-month T-bill rate minus federal funds rate
TB6SMFFM	x_t	Spread: 6-month T-bill rate minus federal funds rate
T1YFFM	x_t	Spread: 1-year treasury rate minus federal funds rate
T5YFFM	x_t	Spread: 5-year treasury rate minus federal funds rate
T10YFFM	x_t	Spread: 10-year treasury rate minus federal funds rate
AAAFFM	x_t	Spread: Aaa corporate yield minus federal funds rate
BAAFFM	x_t	Spread: Baa corporate yield minus federal funds rate
EXSZUSx	$\Delta \log x_t$	Swiss franc to US dollar exchange rate
EXJPUSx	$\Delta \log x_t$	Japanese yen to US dollar exchange rate
EXUSUKx	$\Delta \log x_t$	US dollar to British pound exchange rate
EXCAUSx	$\Delta \log x_t$	Canadian dollar to US dollar exchange rate

Table 18: Prices (20 series)

Series	Transformation	Description
WPSFD49207	$\Delta^2 \log x_t$	Producer price index, finished goods
WPSFD49502	$\Delta^2 \log x_t$	Producer price index, finished consumer goods
WPSID61	$\Delta^2 \log x_t$	Producer price index, intermediate materials
WPSID62	$\Delta^2 \log x_t$	Producer price index, crude materials
OILPRICEx	$\Delta^2 \log x_t$	Crude oil spot price (WTI, spliced series)
PPICMM	$\Delta^2 \log x_t$	Producer price index, metals and metal products
CPIAUCSL	$\Delta^2 \log x_t$	Consumer price index, all items
CPIAPPSL	$\Delta^2 \log x_t$	Consumer price index, apparel
CPITRNSL	$\Delta^2 \log x_t$	Consumer price index, transportation
CPIMEDSL	$\Delta^2 \log x_t$	Consumer price index, medical care
CUSR0000SAC	$\Delta^2 \log x_t$	Consumer price index, commodities
CUSR0000SAD	$\Delta^2 \log x_t$	Consumer price index, durables
CUSR0000SAS	$\Delta^2 \log x_t$	Consumer price index, services
CPIULFSL	$\Delta^2 \log x_t$	Consumer price index, all items excluding food
CUSR0000SA0L2	$\Delta^2 \log x_t$	Consumer price index, all items excluding shelter
CUSR0000SA0L5	$\Delta^2 \log x_t$	Consumer price index, all items excluding medical care
PCEPI	$\Delta^2 \log x_t$	Personal consumption expenditures, chain-type price index
DDURRG3M086SBEA	$\Delta^2 \log x_t$	PCE price index, durable goods
DNDGRG3M086SBEA	$\Delta^2 \log x_t$	PCE price index, nondurable goods
DSERRG3M086SBEA	$\Delta^2 \log x_t$	PCE price index, services

Table 19: Stock market (3 series)

Series	Transformation	Description
`S&P 500`	$\Delta \log x_t$	S&P composite common stock price index
`S&P div yield`	Δx_t	S&P composite stock dividend yield
`S&P PE ratio`	$\Delta \log x_t$	S&P composite stock price–earnings ratio

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